

Design of a 360W isolated ebike charger (AC to DC charger)

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Abstract

This report presents the complete design and validation of a high-efficiency isolated AC–DC battery charger for 24 V, 100 Ah LiFePO₄ battery packs in e-bike and electric mobility applications. The charger accepts universal single-phase AC input from 110–230 V_{rms}, incorporates EMI filtering and full-bridge rectification, and converts power through a half-bridge LLC resonant stage optimized for soft-switching operation and minimal switching losses. The system delivers a regulated 24 V output at 15 A nominal current, achieving 94% efficiency. Charging is controlled via a constant-current/constant-voltage (CC/CV) algorithm coordinated with the battery management system (BMS) to ensure safe, precise, and optimized charging of LiFePO₄ cells. Digital control, real-time monitoring, and fault protection are implemented using an STM32L0 microcontroller, enabling flexible control-loop tuning and advanced protection features. Hardware design, including schematic capture and PCB layout, was developed in Altium Designer, with circuit validation supported by LTspice simulations. Python-based automation tools were used for test execution, data acquisition, and measurement analysis, ensuring repeatable validation and production readiness. The report details measurement techniques, implementation scenarios, and system-level integration for a fully functional, production-ready charger.

This design demonstrates a scalable, high-efficiency charging solution for modern electric mobility systems, combining robust power electronics, intelligent digital control, and manufacturable implementation.

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1 Introduction

This report documents the complete development methodology of a 360 W isolated AC–DC battery charger for 24 V LiFePO₄ battery packs in e-bike and light electric mobility applications. The design process encompasses initial specification, system architecture selection, schematic capture, circuit simulation, PCB layout, experimental validation, and thermal performance characterization.

The power conversion architecture employs a half-bridge LLC resonant topology with galvanic isolation, selected for its high efficiency, soft-switching operation (zero-voltage switching for primary MOSFETs and zero-current switching for secondary rectifiers), and excellent electromagnetic compatibility (EMC) performance. The converter accepts universal single-phase input from 110–230 V_{rms}, input EMI filtering and rectification, and delivers a regulated 20 - 29 V output at 15 A nominal load current with 94% efficiency at the nominal operating voltage.

Battery charging follows a constant-current/constant-voltage (CC/CV) profile, coordinated with the battery management system (BMS) via communication interface. The charger operates in constant-current mode during the bulk charging phase, gradually increasing cell voltage, then transitions to constant-voltage regulation as the battery approaches its target voltage (typically 28.8 V for a 24 V LiFePO₄ pack). This charging strategy ensures optimal charging, and prevents overcharging or thermal runaway conditions.

Digital control, real-time monitoring, and fault protection are implemented using an STM32L0 microcontroller, enabling adaptive control-loop tuning, comprehensive protection features (over-voltage, over-current, over-temperature, short-circuit), and flexible system integration. The hardware design was developed in Altium Designer, with circuit-level validation performed using LTspice simulations, and test automation supported by Python-based data acquisition and analysis tools.

This work presents a scalable charging solution that combines high-efficiency power electronics, intelligent digital control, and robust system-level integration.

2 Design Flow

This report presented the complete engineering development methodology for the 360 W isolated AC–DC e-bike charger, structured to reflect industry-standard power electronics design practices. The workflow demonstrated a systematic approach from initial specification through final validation, showcasing both technical rigor and production readiness.

- **Specification Definition:** Defined electrical specifications, operating conditions, and performance targets including input voltage range (110–230 V_{rms}), output power (360 W), efficiency target ($\geq 94\%$), and safety/regulatory requirements.
- **System Architecture:** Developed the system block diagram selecting LLC resonant topology with galvanic isolation, CC/CV control strategy, and BMS communication interface.
- **Power-Stage Component Selection:** Calculated and selected critical power components including MOSFETs, resonant inductor (L_r), resonant capacitor (C_r), transformer (turns ratio, core selection), output rectifiers, and passive filtering components based on datasheet specifications and application notes.
- **Circuit Simulation:** Performed circuit-level validation using LTspice for transient analysis, soft-switching verification (ZVS/ZCS), efficiency estimation, and stability assessment of the control loop.
- **Schematic and PCB Design:** Captured the schematic and executed PCB layout in Altium Designer, incorporating proper impedance control, thermal management, EMC considerations, and high-voltage isolation creepage/clearance requirements.
- **Embedded Firmware Development:** Implemented digital control loops, BMS communication protocol, real-time monitoring, and fault protection algorithms using STM32L0 microcontroller in STM32 IDE (C/C++).
- **PCB Assembly and Manufacturing:** Generated manufacturing files (Gerber, BOM, pick-and-place), assembled the PCB, and prepared for prototype validation.
- **Prototype Validation:** Conducted comprehensive bench testing including input/output characteriza-

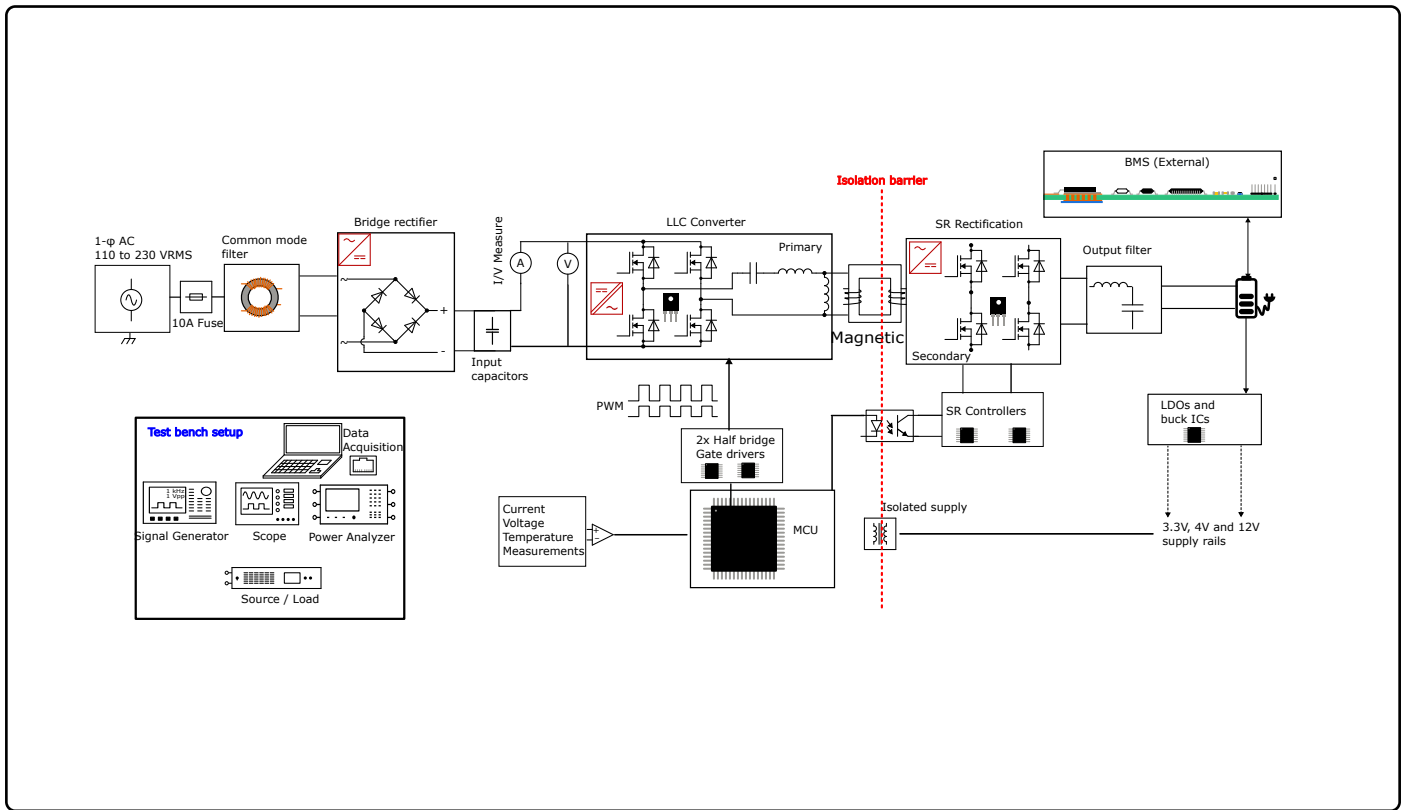


Figure 1: E-bike charger block diagram that was adopted

tion, efficiency measurement, load/transient response, EMC pre-compliance, and thermal performance characterization using calibrated instrumentation.

This structured approach demonstrated not only the final design outcome but also the engineering rationale, design trade-offs, and validation methodology behind each implementation stage—reflecting professional engineering practice suitable for production deployment.

3 Specifications

The following specifications were provided by the client for the 360 W isolated AC–DC e-bike charger. These define the electrical, safety, and performance requirements for the system.

3.0.1 Input Requirements

- **Input voltage:** 110–230 V_{rms} single-phase AC, universal input range.
- **EMI/EMC filtering:** Input-stage EMC filter compliant with IEC 61000-4-6 for conducted emissions reduction.

- **Common-mode filtering:** Common-mode choke for noise attenuation and conducted interference suppression.
- **Protection:** Input fuse protection for over-current and short-circuit safeguarding.
- **Rectification:** Full-bridge diode rectifier for AC-to-DC conversion.
- **Bulk capacitance:** High-voltage bulk capacitor following rectification for DC bus stabilization.
- **Power Factor Correction:** Boost-type PFC stage to achieve power factor ≥ 0.9 and reduce input current harmonics.

3.0.2 Power Conversion Architecture

- **Topology:** Half-bridge LLC resonant converter for isolated power transfer with soft-switching operation (ZVS/ZCS).
- **Primary MOSFETs:** 650 V super-junction MOSFETs in TO-247 package for high-voltage primary switching, optimized for low $R_{DS(on)}$ and fast switching characteristics.
- **Transformer:** High-frequency isolated transformer



optimized for 360 W power transfer, galvanic isolation ≥ 4 kV, and minimal leakage inductance.

- **Secondary rectification:** Synchronous rectification (SR) using SR controller for improved efficiency and reduced conduction losses.
- **Secondary MOSFETs:** 60 V LFPK56 MOSFETs on the secondary rectifier stage, selected for low $R_{DS(on)}$ and compact footprint.

3.0.3 Output and Battery Interface

- **Battery type:** 24 V LiFePO₄ battery pack, 100 Ah capacity.
- **BMS communication:** Digital communication interface with battery management system for CCCV (constant-current/constant-voltage) charging control and safety monitoring.
- **Output voltage:** Regulated 24 V nominal (charging target: 28.8 V for LiFePO₄ pack).
- **Peak output current:** 15 A maximum continuous charging current (360 W nominal power).

3.0.4 Control and Low-Voltage Power Supply

- **Microcontroller:** STM32L0 ARM Cortex-M0+ microcontroller for digital control loop implementation, system monitoring, BMS communication, and fault protection.
- **Voltage rails:** 12 V, 5 V, and 3.3 V supply rails generated from the battery-side DC bus using buck regulators and LDOs for gate drive, analog sensing, and digital logic.
- **Isolation:** Galvanic isolation implemented for high-voltage safety, signal integrity, and compliance with IEC 60950-1/IEC 62368-1 safety standards.

4 Block Diagram

Figure 1 presents the complete system block diagram of the 360 W isolated AC–DC e-bike charger, illustrating the power flow architecture, control subsystems, and interface connections. The design follows a two-stage power conversion approach: (1) an input stage with EMC filtering, rectification, and active boost PFC, and (2) an isolated LLC resonant converter with synchronous rectification.

This architecture was selected to achieve high efficiency, galvanic isolation, and excellent EMC performance while meeting IEC 61000-3-2 harmonic current limits.

4.0.1 Input Stage: EMC Filtering and Rectification

The input stage begins with an **EMI/EMC filter** incorporating a **common-mode choke** to attenuate conducted noise and reduce radiated emissions. This filter serves three critical functions: (1) minimizing radiated and conducted noise on the AC input, (2) reducing the effects of voltage transients from the input source, and (3) protecting the input conductors if a supply failure occurs. The common-mode filter specifically targets conducted noise by presenting high impedance to common-mode currents while allowing differential-mode AC power to pass with minimal attenuation.

Following the EMC filter, a **full-bridge diode rectifier** converts the AC input to pulsating DC. The bridge rectifier is required for AC–DC power supplies and enables universal input operation (110–230 V_{rms}). A **bulk capacitance** following rectification stabilizes the DC bus and filters high-frequency ripple. An inrush current limiter is placed in series with the input to limit the surge current during initial capacitor charging, which can otherwise be $100 \times$ greater than the normal operating current.

4.0.2 Active Power Factor Correction (Boost PFC)

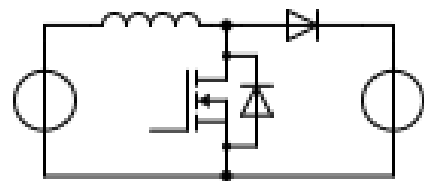


Figure 2: Boost converter topology for the PFC

The boost-type PFC stage is implemented as an active power factor correction circuit to achieve power factor greater than 0.9 and reduce input current harmonics per IEC 61000-3-2. Active PFC brings the power supply's power factor close to 1.0 by shaping the input current waveform to follow the input voltage waveform, thereby minimizing reactive power and harmonic distortion. This stage is critical for meeting regulatory requirements and reducing stress on the AC mains infrastructure.

Why boost PFC over passive PFC?



Passive PFC using inductors can achieve power factor approximately 0.7–0.8 but requires large, heavy magnetic components and provides limited harmonic attenuation. Active boost PFC achieves power factor greater than 0.95 at compact size and provides a regulated high-voltage DC bus (typically 380–400 V) for the subsequent LLC stage. The regulated DC bus also enables the LLC converter to operate with consistent gain characteristics across the universal input range, which is essential for maintaining ZVS conditions. For e-bike chargers requiring compact size and regulatory compliance, active PFC is the industry-standard choice.

4.0.3 Isolated Power Conversion: LLC Resonant Converter

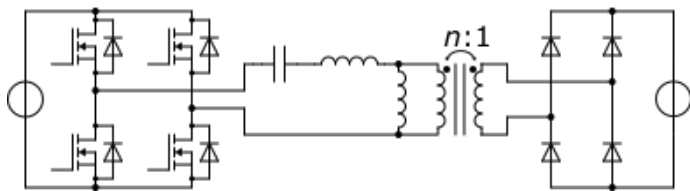


Figure 3: LLC converter topology

The half-bridge LLC resonant converter forms the core power conversion stage, providing isolated power transfer with soft-switching characteristics. The LLC topology was selected over alternative isolated topologies for specific technical reasons.

LLC vs. Hard-Switched Flyback: Flyback converters

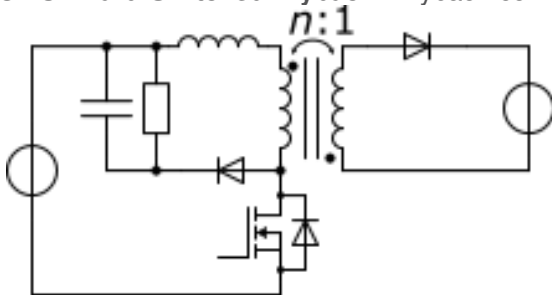


Figure 4: flyback topology

are simple and cost-effective for low-power applications (less than 100 W) but suffer from high switching losses, poor efficiency.

LLC vs. Phase-Shifted Full-Bridge:

Phase-shifted full-bridge converters offer high efficiency

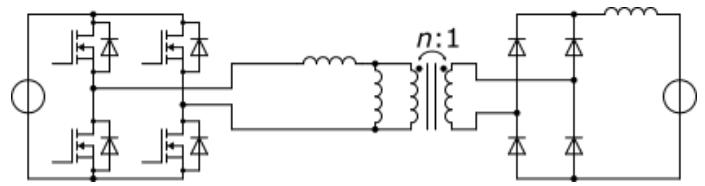


Figure 5: PFSB topology

and power handling but require complex control (four MOSFETs with precise phase timing) and have higher component count. LLC uses only two primary MOSFETs with simpler frequency-control implementation while achieving comparable efficiency ($\geq 94\%$). For 360 W e-bike charger applications, LLC provides the optimal balance of efficiency, complexity, and cost.

The **primary MOSFETs** (650 V super-junction devices in TO-247 package) are optimized for low $R_{DS(on)}$ and fast switching characteristics, enabling efficient high-frequency operation (50–200 kHz). Super-junction MOSFETs provide lower $R_{DS(on)}$. A product compared to conventional MOSFETs, reducing conduction losses.

The **high-frequency isolated transformer** is optimized for 360 W power transfer with galvanic isolation ≥ 4 kV and minimal leakage inductance. The transformer serves as the isolation barrier and provides voltage transformation from the high-voltage DC bus to the 24 V output. Parasitic capacitance between primary and secondary windings is a potential EMI source that must be addressed through proper winding layout and shielding.

4.0.4 Secondary Rectification: Synchronous Rectification (SR)

The **synchronous rectification (SR) stage** uses actively controlled MOSFETs instead of diodes for output rectification, providing:

- **Reduced conduction losses:** MOSFET $R_{DS(on)}$ is significantly lower than diode forward voltage drop V_f
- **Higher efficiency:** SR can achieve 1–2% efficiency improvement over diode rectification at 15 A output
- **Lower thermal stress:** Reduced power dissipation decreases heat generation on the secondary side

Why SR over diode rectification? At 15 A output cur-



rent, a typical Schottky diode with $V_f \approx 0.5$ V dissipates

$$P = V_f \times I = 0.5 \times 15 = 7.5 \text{ W.}$$

Using two diodes in full-wave configuration doubles this to 15 W. In contrast, a 60 V LPAK56 MOSFET with $R_{DS(on)} = 10 \text{ m}\Omega$ dissipates

$$P = I^2 \times R_{DS(on)} = 15^2 \times 0.01 = 2.25 \text{ W.}$$

This represents an 85% reduction in rectification losses, directly contributing to the 94% system efficiency target. SR is essential for high-current, high-efficiency applications.

The **60 V LPAK56 MOSFETs** on the secondary rectifier are selected for low $R_{DS(on)}$ and compact footprint. The LPAK56 package provides excellent thermal performance with low junction-to-case resistance, enabling efficient heat dissipation.

The **SR controller** monitors the transformer secondary voltage and drives the MOSFETs with precise timing to achieve ZCS, minimizing reverse-recovery losses.

4.0.5 Output Filter and Battery Interface

The **output filter** consists of capacitors (and optionally a series inductor for L-filter or Pi-filter configuration) to suppress AC ripple superimposed on the DC output. Without output filtering, the peak-to-peak ripple magnitude would equal the DC voltage, which is unacceptable for battery charging. The filter components are placed close to the battery connection to minimize voltage drop caused by load current variations.

The **battery interface** includes **relays** for safe connection/disconnection of the battery pack. The relays enable fault isolation (over-current, over-voltage, short-circuit) and support the CCCV charging algorithm by controlling current flow during the charging phases. Connection to the **24 V LiFePO₄ battery pack (100 Ah)** is managed through the BMS communication interface.

4.0.6 Control Subsystem: MCU, Gate Driver, and Sensors

The **STM32L0 microcontroller** (ARM Cortex-M0+) implements the digital control loop, system monitoring, BMS

communication, and fault protection algorithms. The STM32L0 was selected for its:

- Low power consumption (suitable for battery-side control power)
- Adequate PWM resolution and frequency for LLC control (up to 32 MHz clock, 16-bit PWM)
- Built-in ADC channels for current/voltage sensing (12-bit, 2.5 MS/s)
- Communication interfaces (UART/I2C) for BMS interface

Why digital control over analog?

Analog control using PWM controllers (e.g., UC384x family) is simpler but lacks flexibility for adaptive control, complex protection algorithms, and BMS communication. Digital control enables:

- (1) adaptive gain tuning for LLC operating across resonant frequency,
- (2) comprehensive fault detection with programmable thresholds,
- (3) firmware updates for field bug fixes, and
- (4) direct BMS communication protocol implementation (e.g., CAN, UART).

For modern e-bike chargers requiring smart features and regulatory compliance, digital control is the preferred approach.

The **gate driver** circuit provides isolated drive signals to the primary MOSFETs, ensuring proper switching timing and protecting the MCU from high-voltage transients. Gate driver isolation is critical for safety and signal integrity in isolated power converters.

Current and voltage sensors (typically Hall-effect sensors or resistive dividers with isolation) provide feedback for the control loop. The current sensor monitors output charging current for CC mode regulation, while the voltage sensor monitors output voltage for CV mode regulation and BMS communication validation.

4.0.7 Low-Voltage Power Supply: Buck Regulators and LDOs

The **low-voltage rails** (12 V, 5 V, and 3.3 V) are generated from the battery-side DC bus using **buck regulators** and



LDOs:

- **12 V rail:** Powers gate drivers, SR controllers, and relay coils
- **5 V rail:** Supplies analog sensing circuits and peripheral interfaces
- **3.3 V rail:** Provides logic power for the STM32L0 MCU

Why buck regulators over linear regulators?

Linear regulators (e.g., 78xx series) dissipate power as heat:

$$P_{\text{loss}} = (V_{\text{in}} - V_{\text{out}}) \times I_{\text{out}}.$$

For 24 V to 3.3 V conversion at 100 mA, a linear regulator dissipates

$$P = (24 - 3.3) \times 0.1 = 2.07 \text{ W},$$

requiring significant heatsinking. Buck regulators achieve 85–95% efficiency, dissipating only 0.1–0.15 W for the same conversion. For battery-side control power where efficiency and thermal management are critical, buck regulators are essential.

Buck regulators provide efficient DC–DC conversion from the higher battery voltage to required logic levels, while LDOs provide clean, low-noise power for sensitive analog and digital circuits.

4.0.8 Thermal and Performance Considerations

The block diagram architecture directly impacts thermal performance and system efficiency:

- **Primary side thermal load:** LLC resonant operation with ZVS minimizes MOSFET switching losses, reducing primary-side heat generation. The TO-247 package provides adequate thermal dissipation for 650 V, 2–3 A primary current.
- **Secondary side thermal load:** SR with ZCS minimizes rectifier losses. The LFPK56 package's low junction-to-case resistance enables efficient heat transfer to the PCB thermal copper.
- **Transformer thermal design:** High-frequency operation reduces transformer size but increases core loss. The transformer is optimized for minimal leakage inductance and appropriate core material (ferrite) to balance efficiency and size.

- **PFC stage thermal load:** The boost PFC dissipates $\approx 2\text{--}3$ W at 360 W output (assuming 97–98% PFC efficiency), requiring adequate copper area and possibly a small heatsink.

Overall system efficiency of 94% at 15 A output corresponds to ≈ 22 W total power dissipation, which must be managed through PCB thermal copper, component placement, and potentially ventilation or forced air if enclosure thermal resistance is high.

4.0.9 Test Instrumentation and Validation Points

The block diagram indicates **test instrumentation connection points** for prototype validation:

- Input AC voltage and current (for power factor and efficiency measurement)
- DC bus voltage (after PFC)
- LLC switching nodes (for ZVS/ZCS verification)
- Output voltage and current (for CC/CV regulation validation)
- Temperature sensors (MOSFETs, transformer, PFC components)

These instrumentation points support the Python-based test automation described in the Design Flow section, enabling automated data acquisition and post-processing for performance validation.

5 Component Selection

This section details the component selection process for the 360 W isolated AC–DC e-bike charger, including mathematical sizing calculations, part requirements, and recommended packages. Components were selected based on datasheet specifications, application notes, and worst-case operating conditions to ensure reliability, efficiency, and regulatory compliance.

5.1 Input Stage

The input stage comprises fuse protection, TVS (transient voltage suppression) diode, common-mode (CM) filter, bridge rectifier, boost PFC stage, bulk capacitors, and current/voltage measurement circuits. Each component is sized and selected as follows.



1. Fuse Protection

Function: Over-current and short-circuit protection for the AC input stage.

Selection Criteria:

- Rated current: $I_{\text{fuse}} \geq 1.5 \times I_{\text{in,max}}$ (for inrush current tolerance)
- Voltage rating: $V_{\text{fuse}} \geq 250 \text{ V}_{\text{rms}}$ (universal input)
- Time-current characteristic: Fast-blow for rapid fault response

Calculation:

Maximum input current at lowest input voltage (110 V_{rms}):

$$\begin{aligned} I_{\text{in,max}} &= \frac{P_{\text{out}}}{V_{\text{in,min}} \times \eta_{\text{total}} \times \text{PF}} \\ &= \frac{360}{110 \times 0.94 \times 0.95} \\ &= 3.68 \text{ A} \end{aligned}$$

where $\eta_{\text{total}} = 94\%$ is total efficiency and $\text{PF} = 0.95$ is power factor.

Fuse rating:

$$I_{\text{fuse}} = 1.5 \times 3.68 = 5.52 \text{ A} \rightarrow \text{Select 6 A fuse}$$

Recommended Part: 6 A, 250 V_{rms}, fast-blow fuse (e.g.,

Littelfuse 0215006.MXL)

Package: Axial leaded or PCB-mount radial fuse (e.g., 5 × 20 mm)

2. TVS (Transient Voltage Suppression) Diode

Function: Protect input stage from voltage transients and surges (e.g., lightning, switching transients).

Selection Criteria:

- Clamping voltage: $V_{\text{clamp}} \geq 1.2 \times V_{\text{in,max}} \times \sqrt{2}$
- Peak pulse power: $P_{\text{pp}} \geq 1000 \text{ W}$ (for IEC 61000-4-5 compliance)
- Breakdown voltage: $V_{\text{BR}} \approx V_{\text{in,max}} \times \sqrt{2}$

Calculation: Maximum AC input voltage: $V_{\text{in,max}} = 230 \text{ V}_{\text{rms}}$

$$V_{\text{peak,max}} = 230\sqrt{2} \approx 325 \text{ V}$$

Clamping voltage:

$$V_{\text{clamp}} = 1.2 \times 325 \approx 390 \text{ V}$$

Recommended Part: 390 V clamping TVS diode (e.g., Littelfuse SA390CA, Bourns CDSF390CA)

Package: SMA (DO-214AC) or SMC (DO-214AA) for high power dissipation.

3. Common-Mode (CM) Filter

Function: Attenuate conducted EMI noise and reduce common-mode currents per IEC 61000-4-6.

Selection Criteria:

- Inductance: $L_{\text{CM}} \geq 10 \text{ mH}$ (for 10–30 MHz attenuation)
- Rated current: $I_{\text{CM}} \geq I_{\text{in,max}} = 3.68 \text{ A}$
- Differential-mode capacitance: $C_{\text{DM}} \leq 4.7$ (to limit leakage current)

Calculation:

For 50 dB attenuation at 150 kHz (EMI start frequency):

$$\begin{aligned} L_{\text{CM}} &= \frac{Z_{\text{source}}}{2\pi f} \times 10^{\frac{\text{attenuation}}{20}} \\ &= \frac{50}{2\pi \times 150 \times 10^3} \times 10^{2.5} \\ &\approx 10 \text{ mH} \end{aligned}$$

where $Z_{\text{source}} = 50 \Omega$ is the standard EMI test impedance.

Recommended Part: 10 mH, 5 A common-mode choke (e.g., TDK DSZV2521A-100M, Würth Electronics 74437346010)

Package: PCB-mount axial or radial choke (e.g., 25 × 21 mm).

4. Bridge Rectifier

Function: Convert AC input to pulsating DC for PFC stage.

Selection Criteria:

- Repetitive peak reverse voltage: $V_{\text{RRM}} \geq 1.2 \times V_{\text{in,max}} \times \sqrt{2}$
- Average forward current: $I_{\text{F(ave)}} \geq 1.5 \times I_{\text{in,max}}$
- Forward voltage drop: $V_{\text{F}} \leq 1.0 \text{ V}$ (for efficiency)
- Reverse recovery time: $t_{\text{rr}} \leq 300 \text{ ns}$ (slow-recovery acceptable for 50/60 Hz)

Calculation:

$$V_{\text{RRM}} = 1.2 \times 230 \times \sqrt{2} = 390 \text{ V} \rightarrow \text{Select 600 V}$$



$$I_{F(ave)} = 1.5 \times 3.68 = 5.52 \text{ A} \rightarrow \text{Select 8 A}$$

Power dissipation at 3.68 A:

$$P_{\text{bridge}} = V_F \times I_{\text{in}} = 0.9 \times 3.68 = 3.31 \text{ W (two diodes in series)}$$

Recommended Part: 600 V, 8 A bridge rectifier (e.g., STMicroelectronics STPSC8H065, Vishay VS-80MW60)
Package: TO-247 or KBU (for 8 A, requires heatsink with $R_{\text{th}} \leq 5 \text{ K/W}$).

5. Boost PFC Stage

Function: Active power factor correction to achieve PF ≥ 0.95 and regulated 390 V DC bus.

5.1 PFC MOSFET

Selection Criteria:

- Drain-source voltage: $V_{\text{DS}} \geq 1.2 \times V_{\text{bus}} = 1.2 \times 390 = 468 \text{ V} \rightarrow \text{Select 600 V}$
- Continuous drain current: $I_{\text{D}} \geq 1.5 \times I_{\text{in,max}}$
- On-resistance: $R_{\text{DS(on)}} \leq 0.5 \Omega$ (for low conduction loss)
- Switching frequency: $f_{\text{sw}} \geq 50 \text{ kHz}$ (CCM operation)

Calculation:

PFC MOSFET RMS current (CCM boost):

$$\begin{aligned} I_{\text{D,RMS}} &\approx I_{\text{in,max}} \times \sqrt{\frac{V_{\text{bus}} - V_{\text{in,peak}}}{V_{\text{bus}}}} \\ &= 3.68 \times \sqrt{\frac{390 - 325}{390}} \\ &= 1.52 \text{ A} \end{aligned}$$

With 1.5×margin: $I_{\text{D}} = 1.5 \times 1.52 = 2.28 \text{ A}$

→ Select 4 A

Conduction loss:

$$\begin{aligned} P_{\text{cond}} &= I_{\text{D,RMS}}^2 \times R_{\text{DS(on)}} \\ &= 1.52^2 \times 0.2 \\ &= 0.46 \text{ W} \end{aligned}$$

$$\text{for } R_{\text{DS(on)}} = 0.2 \Omega$$

Recommended Part: 600 V, 4 A super-junction MOSFET (e.g., Infineon IPW60R045, Onsemi NTH6L020N65SF)
Package: TO-247 or TO-220 (requires heatsink for 0.46 W

+ switching loss).

5.2 Boost Diode

Selection Criteria:

- Reverse voltage: $V_{\text{RRM}} \geq V_{\text{bus}} = 390 \text{ V} \rightarrow \text{Select 600 V}$
- Average forward current: $I_{\text{F(ave)}} \geq I_{\text{out}} = \frac{P_{\text{out}}}{V_{\text{bus}}} = \frac{360}{390} = 0.92 \text{ A}$
- Reverse recovery time: $t_{\text{rr}} \leq 50 \text{ ns}$ (ultrafast/SiC for CCM)
- Forward voltage: $V_F \leq 1.2 \text{ V}$

Calculation: With 2 × margin: $I_{\text{F(ave)}} = 2 \times 0.92 = 1.84 \text{ A} \rightarrow \text{Select 3 A Diode power loss:}$

$$P_{\text{diode}} = V_F \times I_{\text{out}} = 1.0 \times 0.92 = 0.92 \text{ W (SiC diode)}$$

Recommended Part: 600 V, 3 A SiC Schottky diode (e.g., Onsemi CS3D06060L, Wolfspeed C3D06060K).
Package: TO-247 or TO-220 (SiC diodes have low loss, small heatsink sufficient).

5.3 Boost Inductor

Selection Criteria:

- Inductance: L_{boost} for CCM operation at minimum input voltage
- Current rating: $I_{\text{L,max}} \geq I_{\text{in,max}} + \frac{\Delta I_{\text{L}}}{2}$
- Core material: Ferrite (for 50–100 kHz)
- Saturation current: $I_{\text{sat}} \geq 1.2 \times I_{\text{L,max}}$

Calculation:

For CCM at $V_{\text{in,min}} = 110 \text{ V}_{\text{rms}}$, $f_{\text{sw}} = 65 \text{ kHz}$:

$$\begin{aligned} L_{\text{boost}} &= \frac{V_{\text{in,peak}} \times (V_{\text{bus}} - V_{\text{in,peak}})}{f_{\text{sw}} \times P_{\text{out}} \times \Delta I_{\text{L}}} \\ &= \frac{155 \times (390 - 155)}{65 \times 10^3 \times 360 \times 0.2 \times 3.68} \\ &= 280 \mu\text{H} \end{aligned}$$

where $\Delta I_{\text{L}} = 0.2 \times I_{\text{in,max}}$ is 20% ripple current.

Peak inductor current:

$$\begin{aligned} I_{\text{L,max}} &= I_{\text{in,max}} + \frac{\Delta I_{\text{L}}}{2} \\ &= 3.68 + \frac{0.2 \times 3.68}{2} \\ &= 4.05 \text{ A} \end{aligned}$$



With $1.2 \times$ saturation margin:

$$I_{\text{sat}} = 1.2 \times 4.05 = 4.86 \text{ A} \rightarrow \text{Select } 5 \text{ A}$$

Recommended Part: 280 , 5 A ferrite inductor (e.g., PULSE Electronics PLT2815-280K, TDK DSL2520-281K)
Package: Axial or radial shielded inductor (e.g., 28×15 mm).

5.4 PFC Output Capacitor

Selection Criteria:

- Voltage rating: $V_C \geq 1.2 \times V_{\text{bus}} = 468 \text{ V} \rightarrow \text{Select } 500 \text{ V}$
- Capacitance: C_{bus} for hold-up time and ripple voltage
- RMS current rating: $I_{C,\text{RMS}} \geq I_{\text{out}}$
- Low ESR: $\text{ESR} \leq 0.5 \Omega$ (for low ripple loss)

Calculation:

For 10 ms hold-up time at full load:

$$\begin{aligned} C_{\text{bus}} &= \frac{2 \times P_{\text{out}} \times t_{\text{hold}}}{V_{\text{bus}}^2 - V_{\text{min}}^2} \\ &= \frac{2 \times 360 \times 0.01}{390^2 - 300^2} \\ &= 3.3 \mu\text{F} \end{aligned}$$

where $V_{\text{min}} = 300 \text{ V}$ is the minimum PFC output for LLC operation.

Ripple voltage (with $\text{ESR} = 0.3 \Omega$) :

$$\Delta V_{\text{ripple}} = I_{\text{out}} \times \text{ESR} = 0.92 \times 0.3 = 0.28 \text{ V (negligible)}$$

Recommended Part: 500 V, $3.3 \mu\text{F}$ film capacitor (e.g., Vishay 238D335X5500, WIMA WQP335K500)
Package: Radial film capacitor (e.g., 20×35 mm).

6. Bulk Capacitors (After Bridge Rectifier)

Function: Stabilize DC bus before PFC stage and filter high-frequency ripple.

Selection Criteria:

- Voltage rating: $V_{\text{rated}} \geq 1.2 \times V_{\text{in,max}} \times \sqrt{2} = 390 \text{ V} \rightarrow \text{Select } 450 \text{ V}$.
- Capacitance: $C_{\text{bulk}} \geq 1 \mu\text{F/W}$ for ripple filtering.
- RMS current: $I_{C,\text{RMS}} \geq I_{\text{in,max}}$.

Calculation:

$$C_{\text{bulk}} = 1 \mu\text{F/W} \times 360 \text{ W} = 360 \mu\text{F}$$

Select $2 \times 220 \mu\text{F}$

Recommended Part: 450 V, $220 \mu\text{F}$ electrolytic capacitor (e.g., Nichicon UXW450221, Panasonic EEU-FC450221).

Package: Radial electrolytic (e.g., 25×40 mm), use 2 in parallel for current sharing.

7. Current and Voltage Measurement Circuits

7.1 Input Voltage Sensing

Function: Provide scaled AC input voltage feedback to the MCU for PFC control and RMS calculation.

Selection Criteria:

- Voltage divider ratio: $V_{\text{ADC}} = \frac{V_{\text{in,peak}}}{N} \leq 3.3 \text{ V}$ (MCU ADC range).
- Impedance: $R_{\text{total}} \geq 100 \text{ k}\Omega$ (for low power loss).
- Precision: $\leq 1\%$ tolerance resistors.

Calculation:

$$N = \frac{V_{\text{in,peak,max}}}{V_{\text{ADC,max}}} = \frac{325}{3.3} = 98.5 \rightarrow \text{Select } N = 100$$

$$R_1 = 99 \text{ k}\Omega, \quad R_2 = 1 \text{ k}\Omega \quad (\text{voltage divider})$$

Power loss:

$$P_{\text{divider}} = \frac{V_{\text{in,rms}}^2}{R_1 + R_2} = \frac{230^2}{100 \times 10^3} = 0.53 \text{ W}$$

This is acceptable for a 0.5 W to 0.6 W high-voltage resistor string with adequate spacing and thermal margin.

Recommended Part: 99 k Ω , 1 k Ω , 1% tolerance resistors (e.g., Vishay MRS25000, Yageo MFR-25).

Package: SMD 2512 for 0.5 W power rating or axial leaded 0.6 W.

7.2 Input Current Sensing

Function: Measure AC input current for PFC control loop and power calculation.

Selection Criteria:



- Sensing method: Hall-effect sensor for AC, non-invasive, or precision shunt resistor.
- Range: $I_{\text{sense}} \geq I_{\text{in,max}} = 3.68 \text{ A}$.
- Output: Scaled to 0–3.3 V for MCU ADC.

Calculation:

Using a Hall-effect sensor, for example a 5 A range with 250 mV/A output:

$$V_{\text{ADC}} = 3.68 \text{ A} \times 250 \text{ mV/A} = 0.92 \text{ V}$$

This is within the ADC input range.

Recommended Part: 5 A Hall-effect current sensor (e.g., Littelfuse LA55-P, Honeywell CSLA2CD).

Package: PCB-mount through-hole sensor, approximately $20 \times 15 \times 10 \text{ mm}$.

7.3 PFC Output Voltage Sensing

Function: Monitor the 390 V DC bus for PFC regulation and LLC input validation.

Selection Criteria:

- Voltage divider: $N = \frac{390}{3.3} = 118 \rightarrow \text{Select } N = 120$.
- High-voltage resistors: $V_R \geq 200 \text{ V}$ per resistor, using series resistors.

Calculation:

$$R_1 = 119 \text{ k}\Omega \quad (\text{use } 3 \times 40 \text{ k}\Omega \text{ in series}), \quad R_2 = 1 \text{ k}\Omega$$

Recommended Part: 40 k Ω , 200 V, 1% resistors (e.g., Vishay VSR series) + 1 k Ω , 0.25 W.

Package: SMD 2512 or axial leaded 0.6 W.

7.4 Output Current Sensing

Function: Measure 15 A battery charging current for CC mode regulation.

Selection Criteria:

- Sensing method: Low- $R_{\text{DS(on)}}$ MOSFET with current-sense pin or precision resistor.
- Range: $I_{\text{sense}} \geq 15 \text{ A}$.
- Power loss: $P_{\text{sense}} \leq 0.5 \text{ W}$ for efficiency.

Calculation:

Using a precision resistor with $R_{\text{sense}} = 5 \text{ m}\Omega$:

$$P_{\text{sense}} = I_{\text{out}}^2 \times R_{\text{sense}} = 15^2 \times 0.005 = 1.125 \text{ W}$$

This is too high.

Using $R_{\text{sense}} = 2 \text{ m}\Omega$:

$$P_{\text{sense}} = 15^2 \times 0.002 = 0.45 \text{ W}$$

This is acceptable.

Sense voltage:

$$V_{\text{sense}} = 15 \times 0.002 = 30 \text{ mV} \rightarrow \text{Use an op-amp amplifier with gain 1}$$

Recommended Part: 2 m Ω , 5 W precision current-sense resistor (e.g., Vishay WHPT10, Bourns CRSR-5W).
Package: Toggle tab or SMD 2512 for 5 W power rating.

5.2 Primary Power Stage

The primary power stage converts the regulated high-voltage DC bus from the boost PFC stage into a high-frequency, approximately sinusoidal waveform suitable for transfer through the LLC resonant tank and isolation transformer. The stage must support soft switching, low circulating energy, low switching loss, and stable operation over the full universal-input range.

Topology Selection

A half-bridge LLC resonant converter is selected for the primary stage because it provides zero-voltage switching (ZVS) for the primary MOSFETs over most of the operating range, which significantly reduces turn-on losses and dv/dt stress. Compared with hard-switched PWM converters, the LLC topology also reduces electromagnetic interference and enables higher switching frequency operation, improving power density. Compared with a phase-shifted full-bridge, it uses fewer primary switches and a simpler control structure, which is appropriate for a 360 W charger.

Selection Criteria:

- DC bus voltage rating: $V_{\text{DS}} \geq 1.5 \times V_{\text{bus}}$ to cover nominal bus voltage and switching overshoot.
- Current rating: primary switch current must exceed the resonant tank RMS current with thermal margin.



- Switching performance: low gate charge Q_g , low output capacitance C_{oss} , and fast body-diode recovery to support ZVS and low loss.
- Package: thermally robust package with adequate creepage/clearance, preferably TO-247 for prototype and production margin.
- Control compatibility: device must be suitable for high-frequency operation in the 50–200 kHz range, depending on the final resonant design.

Calculations:

For a 360 W output power and assumed full-load efficiency of 94%:

$$P_{in} = \frac{P_{out}}{\eta} = \frac{360}{0.94} = 383 \text{ W}$$

At a nominal PFC bus voltage of 390 V:

$$I_{bus,avg} = \frac{P_{in}}{V_{bus}} = \frac{383}{390} = 0.98 \text{ A}$$

The average bus current is not sufficient for switch sizing because the half-bridge primary devices conduct resonant current, not just average DC bus current. A more relevant design quantity is the resonant tank current, which is typically several times the average bus current depending on tank gain, load, and operating frequency. For conservative sizing, the primary MOSFET RMS current should be chosen with at least a $2\times$ margin over the expected resonant RMS current [ref: LLC design guideline].

A practical voltage margin for the MOSFET is:

$$V_{DS,rated} \geq 1.5 \times 390 = 585 \text{ V}$$

so a 650 V device is the minimum reasonable choice.

If the resonant current RMS is estimated at 2.5 A and the MOSFET on-resistance is 0.25 Ω :

$$P_{cond} = I_{rms}^2 R_{DS(on)} = 2.5^2 \times 0.25 = 1.56 \text{ W}$$

A first-order switching-loss estimate is:

$$P_{sw} \approx \frac{1}{2} V_{DS} I_{sw} (t_r + t_f) f_s$$

where V_{DS} is the drain-source voltage, I_{sw} is the switching current, t_r and t_f are effective rise/fall times, and f_s is the switching frequency. In LLC operation, this loss is reduced substantially because the MOSFETs are intended to switch near zero voltage [ref: ZVS LLC application note].

The resonant frequency is:

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}}$$

and the magnetizing-to-resonant inductance ratio is typically selected in the range:

$$\frac{L_m}{L_r} \approx 3 \text{ to } 8$$

to balance gain range, circulating current, and soft-switching margin [ref: LLC design reference].

Recommended Part: 650 V super-junction MOSFETs with low $R_{DS(on)}$, low Q_g , and low C_{oss} , for example devices in the Infineon, onsemi, ST, or Toshiba 650 V family. For the gate driver, use a half-bridge driver with sufficient high-side bootstrap capability or isolated gate drive, depending on the final control architecture.

Package: TO-247 is recommended for the primary MOSFETs because it offers low thermal resistance, mechanical robustness, and suitable creepage/clearance for a high-voltage prototype. If the design is later optimized for higher power density, a D2PAK or TO-220 variant can be considered, but TO-247 is the safer choice during bring-up and thermal validation.

5.3 Transformer and Resonant Tank

Function: The high-frequency transformer provides galvanic isolation, voltage scaling, and power transfer between the primary LLC stage and the secondary rectification stage. The resonant tank components L_r and C_r set the operating frequency of the converter, shape the gain characteristic, and enable soft-switching operation over the intended load range.

Selection Criteria:

- Isolation rating: transformer primary-to-secondary insulation must satisfy the required safety isolation, typically 4 kV reinforced insulation for prototyping and compliance margin.
- Power rating: core and winding structure must support at least 360 W continuous output without excessive temperature rise.
- Leakage inductance control: transformer leakage inductance should be minimized or intentionally designed as part of the resonant tank, depending on



whether L_r is implemented externally or embedded in the transformer.

- Magnetizing inductance: L_m should be high enough to limit circulating current, but not so high that ZVS is lost at light load.
- Resonant capacitor voltage and current stress: C_r must be a low-loss film capacitor with sufficient RMS current capability.
- Frequency range: the tank should support operation across the required LLC frequency window, typically above and below resonance for gain control.

Transformer Turns Ratio Calculation:

For an output of 24 V nominal and a regulated DC bus of 390 V, the effective turns ratio is selected so that the converter operates near resonance at nominal input and output conditions. For a half-bridge LLC, the primary sees approximately half the bus voltage:

$$V_{\text{pri,eq}} \approx \frac{V_{\text{bus}}}{2} = \frac{390}{2} = 195 \text{ V}$$

A first-order turns ratio estimate is:

$$n = \frac{N_p}{N_s} \approx \frac{V_{\text{pri,eq}}}{V_{\text{out}} + V_d}$$

where V_d accounts for rectifier or synchronous rectifier drops. For a 24 V output and low-loss synchronous rectification, the effective secondary voltage is close to the battery voltage, so:

$$n \approx \frac{195}{24} = 8.1$$

A practical design choice is therefore a turns ratio in the range of 7.5:1 to 8.5:1, depending on the final gain curve and desired operating frequency window.

Resonant Frequency Calculation:

The series resonant frequency of the LLC tank is:

$$f_r = \frac{1}{2\pi\sqrt{L_r C_r}}$$

A suitable design target for this charger is a resonant frequency in the 80 kHz to 150 kHz range, which balances transformer size, switching loss, and EMI performance. For

the purpose of component sizing, choose a nominal resonant frequency of:

$$f_r = 100 \text{ kHz}$$

Resonant Tank Sizing:

The characteristic impedance of the tank is:

$$Z_r = \sqrt{\frac{L_r}{C_r}}$$

A useful first-order approximation for the required tank impedance can be derived from the reflected load resistance. For full-wave rectification with synchronous rectifiers, the reflected load resistance is:

$$R_{\text{ac}} = \frac{8}{\pi^2} n^2 R_{\text{load}}$$

where:

$$R_{\text{load}} = \frac{V_{\text{out}}}{I_{\text{out}}} = \frac{24}{15} = 1.6 \Omega$$

Using $n = 8$:

$$R_{\text{ac}} \approx \frac{8}{\pi^2} \times 8^2 \times 1.6 \approx 83 \Omega$$

A common LLC design approach is to choose the tank characteristic impedance lower than the reflected load resistance so that the converter has adequate gain control range. Selecting:

$$Z_r \approx 12 \Omega \text{ to } 20 \Omega$$

is a reasonable starting point for a 360 W design.

Choosing:

$$Z_r = 15 \Omega, \quad f_r = 100 \text{ kHz}$$

the resonant inductance and capacitance are:

$$L_r = \frac{Z_r}{2\pi f_r} = \frac{15}{2\pi \times 100 \times 10^3} = 23.9 \mu\text{H}$$

$$C_r = \frac{1}{2\pi f_r Z_r} = \frac{1}{2\pi \times 100 \times 10^3 \times 15} = 106 \text{ nF}$$

A practical implementation would therefore use:

$$L_r \approx 24 \mu\text{H}, \quad C_r \approx 100 \text{ nF}$$



with the final values adjusted during simulation and prototype tuning.

Magnetizing Inductance Selection:

The magnetizing inductance is typically chosen as:

$$\frac{L_m}{L_r} \approx 4 \text{ to } 6$$

For $L_r = 24 \mu\text{H}$, a suitable magnetizing inductance is:

$$L_m \approx 120 \mu\text{H}$$

This gives:

$$\frac{L_m}{L_r} = \frac{120}{24} = 5$$

which is a good compromise between gain range and circulating current.

Core Selection:

The transformer core must support the RMS flux density at the selected frequency without saturating. Ferrite is preferred for operation in the 80 kHz to 150 kHz range due to low core loss. The core cross-sectional area and window area should be chosen to accommodate the primary and secondary copper area needed for the required current density, typically 3 A/mm² to 6 A/mm² depending on cooling.

A suitable design should also account for:

- Primary copper loss from RMS current and skin-effect at high frequency.
- Secondary winding loss with synchronous rectification current.
- Interwinding capacitance, which should be minimized to reduce common-mode noise.
- Creepage and clearance between primary and secondary windings.

Recommended Part: Custom-wound ferrite transformer, 360 W class, with approximately 8:1 turns ratio, reinforced insulation, and integrated or external resonant inductance as required by the final design. If leakage inductance is used as part of L_r , the transformer should be

intentionally designed to target the required leakage value during winding construction.

Package: Ferrite transformer on an E- or ER-type core with bobbin winding, reinforced insulation tape, and PCB-mounted or leaded termination depending on the final mechanical design. The resonant capacitor should be a polypropylene film capacitor with low ESR and high RMS current capability, placed close to the primary MOSFET half-bridge to minimize loop inductance.

Implementation Notes The final values of L_r and C_r should be validated in LTspice and then refined on hardware by measuring resonant frequency, gain curve, and ZVS margin. Small deviations from the calculated values are expected because transformer parasitics, diode capacitance, and PCB stray inductance all influence the effective tank parameters.

5.4 Secondary power stage

The synchronous rectifier uses 60V LPAK56 MOSFETs chosen for low conduction loss and compact package size. The SR controller manages timing to reduce diode-like losses on the secondary side and improve thermal performance. This is especially useful in a 24V, 15A charger where secondary conduction losses become significant.

Function: The secondary rectification stage converts the high-frequency AC waveform from the transformer secondary into regulated DC for battery charging. In this design, synchronous rectification is used instead of diode rectification to reduce conduction loss, improve efficiency, and minimize thermal stress at the 15 A output current.

Topology Selection:

Synchronous rectification is preferred over Schottky diodes because the low output voltage makes diode forward-drop losses significant. At 24 V and 15 A, even a modest diode drop would dissipate several watts per device, which would directly reduce charger efficiency and increase heatsinking requirements. MOSFET-based rectification allows the effective forward drop to be replaced by $I^2 R_{DS(on)}$, which is substantially lower at this current level.



Selection Criteria:

- Voltage rating: secondary MOSFET $V_{DS} \geq 2 \times V_{out}$ with margin for ringing and leakage spikes.
- Current rating: each rectifier MOSFET must support the full secondary current pulse with adequate thermal margin.
- Low conduction loss: minimum $R_{DS(on)}$ to reduce I^2R dissipation.
- Fast body diode and low reverse recovery charge to improve commutation behavior.
- Low gate charge Q_g for efficient drive and reduced controller burden.
- Package thermal performance: large exposed pad package preferred to spread heat into the PCB copper.

Calculation:

For a 24 V, 15 A output, the rectifier stage must support approximately:

$$P_{out} = 24 \times 15 = 360 \text{ W}$$

A Schottky diode rectifier with a typical forward voltage of 0.5 V would dissipate approximately:

$$P_{diode} = V_F I_{out} = 0.5 \times 15 = 7.5 \text{ W}$$

per conducting diode, and roughly twice that in a full-wave implementation depending on the current path. This level of loss is undesirable for a compact charger.

Using a synchronous rectifier MOSFET with $R_{DS(on)} = 10 \text{ m}\Omega$:

$$P_{SR} = I_{out}^2 R_{DS(on)} = 15^2 \times 0.01 = 2.25 \text{ W}$$

This is a major reduction relative to diode rectification and is more consistent with the 94% system efficiency target.

If the current is shared between two SR devices, the effective loss per device can be estimated from the conduction duty cycle. For example, assuming each device conducts for half the period and the RMS current per device is approximately 10.6 A:

$$P_{device} \approx 10.6^2 \times 0.01 = 1.12 \text{ W}$$

per MOSFET, which is manageable with proper PCB copper area.

Voltage Rating Margin:

Although the output voltage is only 24 V nominal, secondary-side MOSFETs must withstand leakage spikes, switching transients, and any resonant overshoot during commutation. A 60 V MOSFET is therefore the standard choice:

$$V_{DS, rated} \geq 60 \text{ V}$$

This provides ample margin above the battery voltage and secondary-side transient conditions.

Gate Drive Requirements:

The synchronous rectifier must be timed so that the MOSFET channel conducts only when current is flowing in the correct direction. The controller should avoid cross-conduction and reverse current flow during commutation. Important requirements include:

- Accurate timing relative to transformer secondary voltage polarity.
- Dead-time insertion to prevent shoot-through.
- Fast turn-off to avoid reverse conduction.
- Stable operation over light-load and full-load conditions.

Recommended Part: A 60 V low- $R_{DS(on)}$ MOSFET in an LFPK56 or equivalent compact power package is recommended for the synchronous rectifier. Devices in the 5 m Ω to 15 m Ω class are suitable, depending on thermal constraints and gate-drive capability.

Package:

LFPK56 is preferred because it provides:

- Very low thermal resistance.
- Excellent current handling for compact layouts.
- Low parasitic inductance.
- Strong compatibility with high-density PCB copper heat spreading.

If the thermal design is less constrained, D2PAK or TO-220 alternatives can also be used, but LFPK56 is the best fit



for a compact, high-efficiency charger.

Implementation Notes:

The secondary rectification loop should be laid out with the shortest possible current path between the transformer secondary, MOSFETs, and output capacitors. This minimizes parasitic inductance and reduces ringing at turn-off. Output capacitors should be placed close to the rectifier to absorb commutation ripple and to reduce voltage droop during switching transitions. A dedicated synchronous rectifier controller is recommended to detect current polarity and optimize timing automatically.

5.5 Battery Interface and Relay Control

Function: The battery interface provides a safe, controlled connection between the charger output and the LiFePO_4 battery pack. It must support controlled connection and disconnection during normal charging, fault isolation during over-voltage or over-current events, and a predictable precharge sequence to limit inrush current into the output capacitors and battery-side wiring. The relay system also prevents uncontrolled arcing at the connector and allows the charger to enter a defined safe state when the battery is removed or the BMS requests shutdown.

Selection Criteria:

- Contact current rating: relay contacts must support at least the maximum continuous charge current with thermal margin.
- Voltage rating: contacts must withstand the maximum battery-side DC voltage and switching transients.
- Coil voltage: must match the available low-voltage control rail, typically 12 V or 5 V.
- Coil power: should be low enough to avoid unnecessary auxiliary dissipation.
- Contact resistance: low enough to minimize I^2R loss and voltage drop at 15 A.
- Isolation: coil-to-contact isolation must satisfy safety and noise-immunity requirements.
- Switching endurance: adequate mechanical and electrical life for repeated charge cycles.
- Control logic compatibility: relay driver must interface cleanly with MCU GPIO or a dedicated low-side driver stage.

Relay Topology:

A practical battery interface for a 24 V charger is a main output relay in series with the positive battery lead, combined with a precharge path in parallel. The main relay is normally open during startup. The precharge path, consisting of a resistor and a smaller relay or transistor-controlled switch, charges the output capacitors before the main relay closes. This reduces contact stress and prevents a high inrush pulse into the battery-side filter network.

For safety-critical designs, the output disconnect function may be implemented with two relays in series, one on the positive conductor and one on the return conductor, or with a main contactor plus precharge relay depending on the mechanical layout and fault philosophy. A dual-relay arrangement improves fault isolation because a single welded contact cannot leave the pack permanently connected.

Calculation:

At 15 A charging current, the relay contact dissipation is:

$$P_{\text{contact}} = I^2 R_{\text{contact}}$$

If the relay contact resistance is 50 m Ω , then:

$$P_{\text{contact}} = 15^2 \times 0.05 = 11.25 \text{ W}$$

This is too high for a compact output relay, so a lower-resistance power relay or contactor with a contact resistance in the low milliohm range is preferred. If the contact resistance is 5 m Ω :

$$P_{\text{contact}} = 15^2 \times 0.005 = 1.125 \text{ W}$$

which is much more acceptable for continuous operation.

Precharge Resistor Calculation:

To limit the inrush current into the output capacitor bank, a precharge resistor is placed in series before the main relay closes. If the battery-side DC bus is 28.8 V maximum and the desired precharge current is limited to 0.5 A:

$$R_{\text{pre}} = \frac{V}{I} = \frac{28.8}{0.5} = 57.6 \text{ } \Omega$$

A standard value of 56 Ω or 68 Ω is suitable depending on the required precharge speed. The resistor power during



precharge is:

$$P_{pre} = \frac{V^2}{R} = \frac{28.8^2}{56} = 14.8 \text{ W}$$

so a resistor with at least 25 W pulse capability is recommended, even though the duty cycle is low.

Control Strategy: The MCU controls the relays using a sequenced state machine. The recommended order is:

- Verify battery presence, pack voltage, and BMS permissive status.
- Energize the precharge path and wait until output capacitor voltage approaches battery voltage.
- Close the main relay once the voltage difference is below the selected threshold.
- Disable the precharge path after the main relay is fully closed.
- Continuously monitor output current, output voltage, relay state, and fault inputs during charging.
- Open the main relay immediately on over-voltage, over-current, over-temperature, communication loss, or BMS fault.

Driver Requirements:

The relay coil should not be driven directly from the MCU pin. A low-side N-channel MOSFET or BJT driver stage is preferred, with a flyback diode across the coil to clamp turn-off transients. If the coil is inductive and the switching frequency is low, the driver dissipation is small, but the flyback network is still mandatory to protect the control electronics. A base or gate resistor should be included to limit inrush into the driver device and improve switching stability.

BMS and Interlock Logic:

The relay control must be tied to the BMS permissive signal and charger fault logic. The relay should remain open unless all of the following are true:

- Battery pack voltage is within the safe charging window.
- BMS indicates that charging is allowed.
- No over-voltage, over-current, or over-temperature fault is active.
- The MCU watchdog and control firmware are running normally.

If any permissive signal is lost, the MCU must open the relay and force the converter into a safe shutdown state.

Recommended Part: A PCB-mount power relay or DC contactor rated for at least 30 A continuous current and 60 V DC contact switching, with low contact resistance and a 5 V or 12 V coil depending on the auxiliary rail. For the precharge element, use a wirewound or metal-oxide resistor with pulse capability appropriate for the calculated energy. If a compact solution is preferred, a sealed automotive-style relay or DC-rated contactor is preferable to a small signal relay because the charger current is far above the safe range of general-purpose PCB relays.

Package: Sealed power relay or DC contactor with screw terminals or high-current PCB pins, mounted with adequate creepage, clearance, and thermal spacing from low-voltage control circuitry. The precharge resistor should be mounted to allow heat dissipation and kept away from temperature-sensitive components.

Implementation Notes:

The relay coil driver should include a dedicated flyback diode, local decoupling, and, if needed, a transistorized enable stage so the MCU can cleanly disable the output in firmware. Relay contact routing should minimize loop area and resistive loss, and the precharge branch should be physically separated from the main current path to simplify fault diagnosis. The charger output should not energize the battery connector until the output capacitors have been precharged and the BMS has authorized charging.

The battery interface should use a main power relay or contactor with a controlled precharge path and MCU-driven coil logic. This arrangement reduces inrush stress, improves connector safety, and provides a deterministic means of disconnecting the battery under fault conditions.

5.6 Control Board Component Selection

The control board provides closed-loop regulation, protection logic, measurement acquisition, synchronous rectifier timing, and gate-drive supervision for the charger. It must therefore combine deterministic digital timing, low-noise analog sensing, robust peripheral interfaces, and stable



auxiliary power generation. The architecture implemented here uses an STM32L053C8T6 microcontroller as the central control device, Texas Instruments synchronous rectifier controllers, TI buck regulators, and low-noise TI LDOs for the internal rails.

5.6.1 Microcontroller Selection

The microcontroller implements the digital control layer for the charger. Its tasks include ADC acquisition of input and output voltage/current feedback, PWM generation for frequency control of the LLC power stage, fault management, communication with the battery-management system, soft-start sequencing, telemetry, and service/debug access.

Selection Criteria:

- Cortex-M0+ core with deterministic interrupt latency for control tasks.
- Multiple ADC channels for simultaneous sensing of input voltage, input current, output voltage, output current, and thermal inputs.
- Timer resources for PWM, frequency modulation, dead-time insertion, and capture/compare timing.
- UART, I²C, and SPI support for debug, BMS communication, and peripheral expansion.
- Adequate Flash and RAM for closed-loop firmware, calibration tables, protection state machines, and event logging.
- Low-power operating modes to reduce standby consumption.

Selected Part: **STM32L053C8T6**.

Why this part: The STM32L053C8T6 belongs to the STM32L0 family and provides a low-power Cortex-M0+ core with sufficient timer, ADC, and serial interface resources for a compact power-supply controller. Its 32 MHz operating capability is adequate for control-loop computation, real-time monitoring, and frequency-based LLC control, while its low-power features reduce auxiliary consumption during standby and fault states.

Evaluation Board: NUCLEO-L053R8 is a practical development and firmware-validation board for this family, it was useful for control-loop bring-up, ADC calibration,

and peripheral validation before custom PCB deployment.

Sensing and Control Logic:

The MCU receives scaled feedback signals from the input and output sensing circuits. The sensed AC input voltage is rectified or sampled synchronously and converted by the ADC for RMS estimation, mains supervision, and PFC control. Input current feedback is used for average current-mode or peak-current supervision, depending on the final control algorithm. Output voltage feedback is used to regulate the LLC frequency command, while output current feedback enforces constant-current charging during the bulk phase and constant-voltage charging near the end of charge.

The ADC results are processed in firmware using gain and offset calibration coefficients.

The firmware computes the error between the measured quantity and the reference setpoint:

$$e(t) = x_{\text{ref}} - x_{\text{meas}}$$

where x_{meas} is the sensed voltage, current, or derived RMS quantity. The error is then passed through the corresponding control law, typically PI or gain-scheduled PI for LLC frequency regulation. For the LLC stage, the control output is not duty cycle in the conventional PWM sense, but switching frequency:

$$f_{\text{sw}} = f_r + \Delta f(e)$$

where f_r is the resonant frequency and $\Delta f(e)$ is the frequency correction generated by the control loop.

PWM and Duty Logic:

For the half-bridge LLC primary stage, the MCU does not vary duty cycle as in a fixed-frequency PWM converter. Instead, it generates complementary gate-drive signals with fixed 50% duty per leg and inserts dead time to prevent shoot-through. Output power is controlled by varying the switching frequency around resonance. When the measured output voltage is below target, the firmware reduces the switching frequency toward resonance or below resonance to increase tank gain. When the output voltage rises above target, the firmware increases switching frequency above resonance to reduce gain and circulating current.



A practical timer configuration is:

- Complementary outputs with programmable dead time for the half-bridge driver.
- Frequency-modulated timer period for LLC control.
- ADC triggering aligned to the PWM cycle for low-noise synchronous sampling.
- Interrupt-based fault shutdown for over-voltage, over-current, over-temperature, and loss-of-feedback events.

For synchronous rectification, the MCU may either supervise a dedicated SR controller or provide enable/disable coordination with the charging state machine. In this design, the SR timing itself is preferably handled by a TI SR controller, while the MCU handles higher-level sequencing and fault responses.

5.6.2 Synchronous Rectifier Controller

The synchronous rectifier controller drives the secondary-side MOSFETs to reduce rectification loss and improve efficiency at 15 A output current. It senses the transformer secondary polarity and ensures that the MOSFET channel turns on only when forward current is present.

Selection Criteria:

- Compatible with 24 V output systems and LLC synchronous rectification.
- Drain-source sensing to detect conduction polarity without a separate current transformer.
- High-speed turn-on and turn-off with blanking to suppress false triggering.
- Interlock function to prevent cross-conduction between the two SR MOSFETs.
- Supply and voltage ratings compatible with the secondary-side bias rail.

Calculation:

At 15 A output current, diode rectification is too lossy. Using a representative 0.5 V diode drop:

$$P_{\text{diode}} = 0.5 \times 15 = 7.5 \text{ W}$$

per conducting path. With a 10 mΩ MOSFET:

$$P_{\text{SR}} = 15^2 \times 0.01 = 2.25 \text{ W}$$

This makes SR the correct choice for the secondary stage.

Selected Part: Texas Instruments UCC24624.

The UCC24624 is suitable for LLC resonant converter secondary rectification and uses drain-to-source voltage sensing with adaptive proportional gate drive. It includes blanking intervals, interlock logic, and voltage-sensing capability appropriate for low-voltage, high-current outputs such as 24 V battery chargers.

Package: SOIC or small-outline power-control package with short gate and sense routing.

5.6.3 Primary Gate Driver

The gate driver drives the half-bridge MOSFETs on the primary side, supplying the peak source and sink current needed for fast transitions at high frequency. It also enforces dead time and protects the power switches from shoot-through.

Selection Criteria:

- Half-bridge support with high-side and low-side drive.
- Strong gate-drive current capability for low total transition loss.
- UVLO to prevent partial enhancement at insufficient supply voltage.
- High dv/dt immunity for operation on a 390 V switching node.
- Compatibility with 10 V to 12 V gate-drive rail.

Calculation:

If one MOSFET has total gate charge $Q_g = 70 \text{ nC}$, gate voltage $V_{GS} = 12 \text{ V}$, and switching frequency $f_s = 100 \text{ kHz}$, the gate-drive power per device is:

$$P_{\text{gate}} = Q_g V_{GS} f_s = 70 \times 10^{-9} \times 12 \times 100 \times 10^3 = 0.084 \text{ W}$$

This is small individually, but the driver and supply rails must still be sized for transient current and noise immunity.

Recommended Part: A TI half-bridge gate-driver family part suitable for resonant converters and high dv/dt operation.

Package: SOIC-8 or QFN with local bootstrap and decoupling capacitors placed close to the device pins.



5.6.4 Low-Voltage Regulators

The auxiliary supply generates the board rails used by the MCU, SR controller, gate driver, and sensing analog front-end. The preferred architecture uses a TI synchronous buck regulator for bulk step-down conversion and a TI LDO for low-noise regulation.

Selection Criteria:

- Buck regulator input range compatible with the available auxiliary supply.
- Output current sufficient for MCU, SR controller, driver bias, and sensor loads.
- High conversion efficiency to minimize heating in the sealed control enclosure.
- LDO output noise low enough for ADC and timing-sensitive analog circuitry.
- Good transient response during startup, fault recovery, and state transitions.

Calculation:

If the board requires 12 V at 150 mA from a 24 V input rail:

$$P_{\text{out}} = 12 \times 0.15 = 1.8 \text{ W}$$

At 90% efficiency:

$$P_{\text{loss}} = \frac{1.8}{0.9} - 1.8 = 0.2 \text{ W}$$

A 12 V to 5 V or 12 V to 3.3 V LDO stage is then used only for the low-current, low-noise rails where linear dissipation is acceptable. For example, 5 V to 3.3 V at 80 mA dissipates:

$$P_{\text{LDO}} = (5 - 3.3) \times 0.08 = 0.136 \text{ W}$$

Selected Parts:

For the buck stage, a TI synchronous buck regulator such as TPS54202 is a suitable class of device for intermediate rails, while a higher-integration option such as TPSM63603 can be used when a compact power-module implementation is preferred. For the LDO stage, TPS7A44 is attractive because it supports a wide input range and can provide a clean bias rail with low quiescent current.

Package: Use QFN or HTSSOP for the buck regulator and SOT-223, SOT-23, or DFN for the LDO, depending on current and thermal requirements.

5.6.5 Interface and Programming

The control board should expose programming and debug interfaces for firmware bring-up, calibration, and fault analysis.

Selection Criteria:

- SWD interface for programming and debugging.
- UART for bootloader or field logging.
- I²C for sensors and support ICs.
- Test points for all regulated rails and main feedback nodes.

Recommended Part: Standard 0.05-inch or 0.1-inch debug header footprint with labeled pins for SWDIO, SWCLK, NRST, TX, RX, GND, and target voltage.

Package: Board-edge header or low-profile debug connector.

6 Design creation and Simulation

The LLC converter was first developed and verified in LTspice to validate resonant tank behavior, switching transitions, output ripple, rectifier conduction, and the influence of non-ideal parasitics before hardware implementation.

The simulation workflow was used to confirm that the selected resonant parameters, transformer ratio, and semiconductor devices could meet the 360 W output target while preserving zero-voltage switching and acceptable current stress across the intended operating range.

6.0.1 Simulation Procedure

The simulation was built in stages rather than as a single full-system model. The first stage modeled only the LLC resonant tank, half-bridge excitation, and transformer equivalent circuit to confirm the approximate resonant frequency, current shape, and gain behavior. In this stage, the resonant inductor L_r , resonant capacitor C_r , magnetizing inductance L_m , transformer turns ratio, and bus voltage were defined using the nominal design values derived from the hand calculations.

The second stage added the primary MOSFETs, gate-drive timing, and dead-time interval to check that both switches turned on under near-zero drain-source voltage. This was essential for verifying zero-voltage switching over the expected load window. The switching frequency was

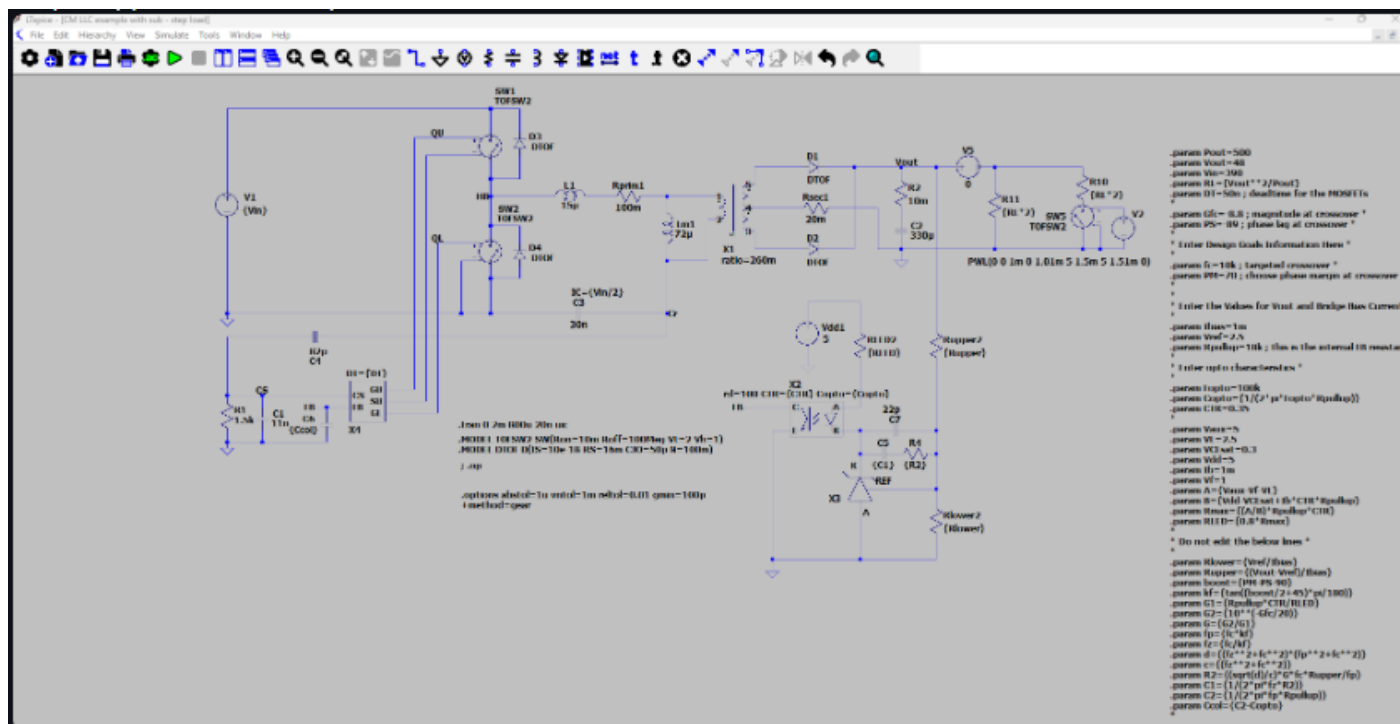


Figure 6: LTspice simulation model used for LLC converter verification.

then swept around resonance to observe the gain characteristic and determine the operating region required for constant-current and constant-voltage charging.

The third stage added the secondary-side synchronous rectification stage, output capacitor bank, and resistive load. This allowed the output ripple, rectifier current waveform, and secondary conduction timing to be evaluated under light-load, nominal-load, and full-load conditions. Finally, non-ideal elements such as winding resistance, capacitor ESR, MOSFET output capacitance, leakage inductance, and parasitic capacitances were introduced to make the model more representative of the final hardware.

6.0.2 Parasitic Modeling

Parasitic components were included because they strongly affect LLC converter performance at high frequency. The transformer winding resistance increases conduction loss and reduces efficiency, while leakage inductance alters the effective resonant tank value and shifts the resonant frequency. The MOSFET output capacitance contributes additional resonant energy that affects turn-on conditions and switching loss. Capacitor ESR and inductor series resistance also modify the Q-factor of the tank and influence output ripple.

The final simulation therefore included the following parasitics:

- Transformer primary and secondary winding resistance.
- Leakage inductance as part of, or in addition to, the resonant inductance.
- MOSFET output capacitance and body-diode behavior.
- Resonant capacitor ESR.
- Output capacitor ESR and equivalent series inductance.
- PCB trace resistance and small loop inductance in the switching paths.

Including these non-idealities allowed the simulated waveforms to better match the expected prototype behavior and reduced the risk of large deviations during hardware bring-up.

6.0.3 Key Simulation Checks

The simulation was used to verify several critical design objectives:

- Resonant frequency agreement between the analytical design and the LTspice model.



- Zero-voltage switching on the primary MOSFETs at nominal and near-nominal load.
- Acceptable secondary synchronous rectifier timing and low reverse conduction.
- Output ripple magnitude within the battery-charging requirement.
- Stable gain transition between constant-current and constant-voltage operating regions.
- Reasonable current stress in the primary switches, resonant tank, and secondary rectifiers.

6.0.4 Final Design Results

The final LTspice model confirmed that the chosen LLC parameters were appropriate for the 360 W charger. With the selected resonant tank values, the simulated converter operated near resonance at the nominal bus voltage and output load, producing the expected high-frequency waveform at the transformer primary and maintaining soft-switching behavior over the intended operating range.

At nominal operating point, the primary drain-source waveforms showed that the MOSFETs were turned on after the switch-node voltage had fallen close to zero, confirming ZVS. The secondary rectifier currents were also well aligned with the transformer secondary voltage polarity, reducing unnecessary diode conduction and supporting high efficiency. Output ripple remained low enough for battery charging, and the current waveforms remained smooth enough to avoid excessive stress on the transformer and rectifier devices.

The final model also showed that the inclusion of parasitics slightly shifted the resonant frequency downward relative to the ideal analytical value, which is expected in a practical implementation. This shift was compensated by tuning the nominal switching frequency window in firmware and by confirming that the final frequency range still preserved gain control margin. The simulation therefore validated both the selected hardware topology and the control strategy before PCB fabrication.

6.0.5 Design Interpretation

The LTspice study confirmed that the converter is best operated with frequency control rather than duty-cycle modulation. The duty ratio of the half-bridge remains approximately fixed, while regulation is achieved by moving the switching frequency relative to resonance. This behavior is

consistent with LLC design theory and provides the main mechanism for output regulation, soft switching, and load adaptation.

Overall, the simulation stage demonstrated that the selected resonant tank, transformer ratio, MOSFETs, synchronous rectification strategy, and output filtering network form a technically coherent design. The resulting waveforms and stress levels supported the transition from schematic design to PCB implementation and prototype validation.

7 Schematic and PCB Design

The complete electrical implementation of the charger was captured in Altium Designer, beginning with the input front end, followed by the boost PFC stage, the isolated LLC power stage, the synchronous rectifier, the control circuitry, and the auxiliary low-voltage supplies. The schematic was partitioned into functional blocks to improve readability, simplify net management, and allow independent verification of each subsystem before integration. The PCB layout was then implemented as a compact two-layer design with careful attention to creepage, clearance, thermal dissipation, current density, and noise-sensitive routing.

7.0.1 Schematic Capture Method

The schematic was developed using a hierarchical structure so that each functional block could be validated independently before system integration. The input protection and EMI network were placed at the front of the design, followed by rectification and PFC conversion, then the LLC resonant power stage, synchronous rectification, control circuitry, sensing circuits, and auxiliary power rails. This approach reduces schematic complexity, improves debug efficiency, and makes it easier to trace power flow and signal flow across the design.

Critical nets such as the AC input, PFC bus, half-bridge switching node, transformer primary, secondary rectified output, and control references were clearly separated and labeled. Net naming was kept consistent with the design calculations so that the schematic, simulation model, and PCB layout could be cross-checked without ambiguity. Power ports and ground references were managed carefully to avoid accidental shorting between the high-voltage primary domain and the isolated secondary/control

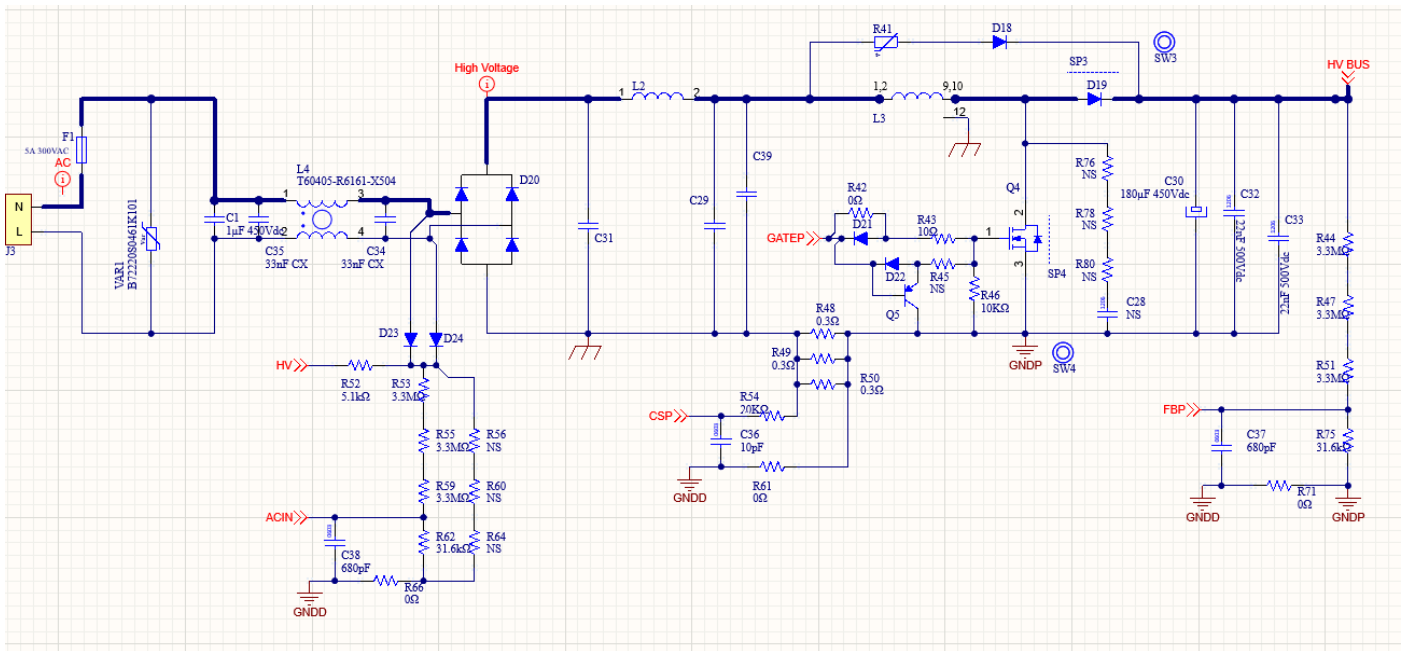


Figure 7: Altium design of the input front end: connectors, EMI filter, bridge rectifier, and boost PFC.

domain.

7.0.2 PCB Design Rules

The PCB was designed as a two-layer board with top-layer component placement optimized for short power loops and bottom-layer copper used as a return path and thermal spreading layer. The main routing rules followed during layout were:

- Maintain sufficient creepage and clearance between the primary side, secondary side, and low-voltage control area.
- Keep high-current paths as short and wide as possible to minimize copper loss and voltage drop.
- Minimize the loop area of switching nodes to reduce radiated and conducted EMI.
- Place decoupling capacitors directly adjacent to the pins of power ICs, gate drivers, and control ICs.
- Separate noisy power ground paths from sensitive analog ground returns and join them at a controlled star point.
- Use thermal vias and large copper pours under devices with significant power dissipation.
- Avoid routing feedback and sensing traces near high dv/dt nodes such as the half-bridge midpoint and transformer primary connections.

The PCB layout was developed with a strong focus on power integrity, EMI control, and safety isolation. The layout maintains 3 mm clearance between neutral and live conductors on the primary side, and 6 mm clearance between the primary and secondary domains to support the required isolation barrier and reduce the risk of electrical breakdown. High-frequency switching loops were kept as compact as possible to minimize parasitic inductance, switching overshoot, and radiated emissions, while sensitive measurement, control, and communication traces were routed away from noisy power nodes.

The primary and secondary sections were physically separated to preserve the isolation barrier, with creepage and clearance considered throughout the board layout and transformer interface. Thermal performance was addressed by spreading losses across the board using copper for heat dissipation, and by positioning major power devices to simplify cooling and reduce local hot spots. This layout approach improves reliability, reduces EMI susceptibility, and supports safe operation under high-voltage conditions.

7.0.3 Input Front-End Layout

As shown in Figure 7, the input front end contains the connectors, EMI filter, bridge rectifier, and boost PFC stage.

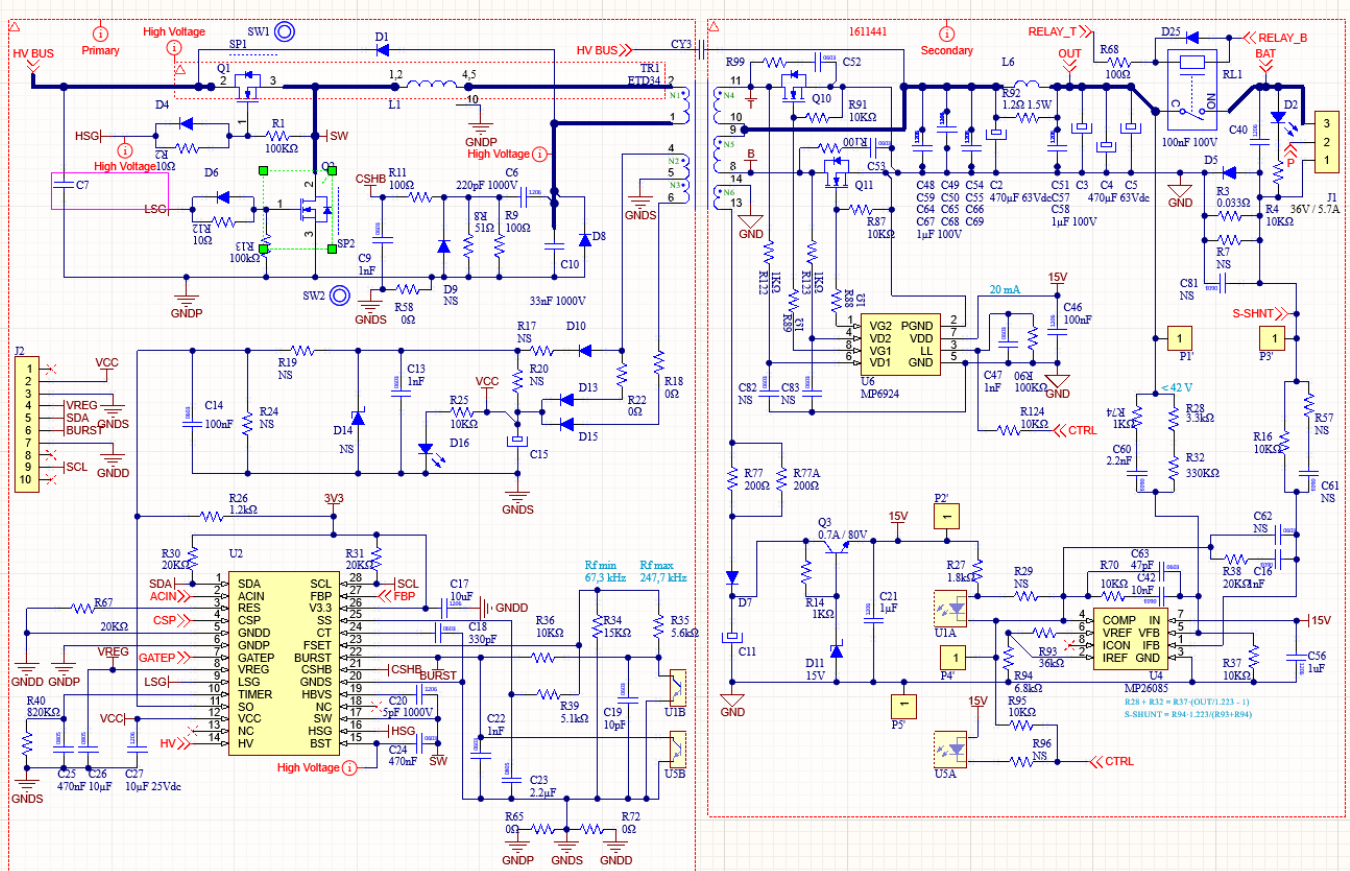


Figure 8: Altium design of the LLC converter and its magnetics.

These components were placed in the order of power flow to keep the high-current AC and DC paths compact. The EMI filter was positioned immediately after the input connector so that conducted-noise attenuation occurred before the rectifier and PFC stage. The bridge rectifier and boost switch path were then arranged to minimize the current loop formed by the rectifier, boost inductor, switching MOSFET, diode, and bus capacitor.

The boost inductor was placed close to the switching node and bus capacitor to reduce parasitic inductance in the high di/dt path. The PFC current-sense elements were routed with Kelvin-style connections to avoid contamination from power-ground voltage drop. The DC bus capacitors were positioned close to the boost output to limit ripple-current loop area and improve transient behavior. Thermal copper was added beneath the bridge rectifier and PFC MOSFETs, with additional vias to transfer heat to the opposite layer.

7.0.4 LLC and SR Layout

Figure 8 shows the isolated LLC converter and its magnetics. The LLC resonant tank was placed as close as possible to the primary half-bridge MOSFETs and transformer primary pins in order to minimize loop inductance and improve ZVS performance. The resonant capacitor was placed adjacent to the switching devices, while the resonant inductor and transformer were arranged to reduce current path length and preserve the intended resonant behavior.

The transformer was oriented so that the primary and secondary current paths were well separated on the board, with the secondary rectification devices placed near the transformer secondary pins to minimize the rectifier commutation loop. This is especially important because synchronous rectification depends on low parasitic inductance and accurate switching timing. The SR controller sensing pins were routed away from the switching node and coupled to the transformer secondary with short, low-noise

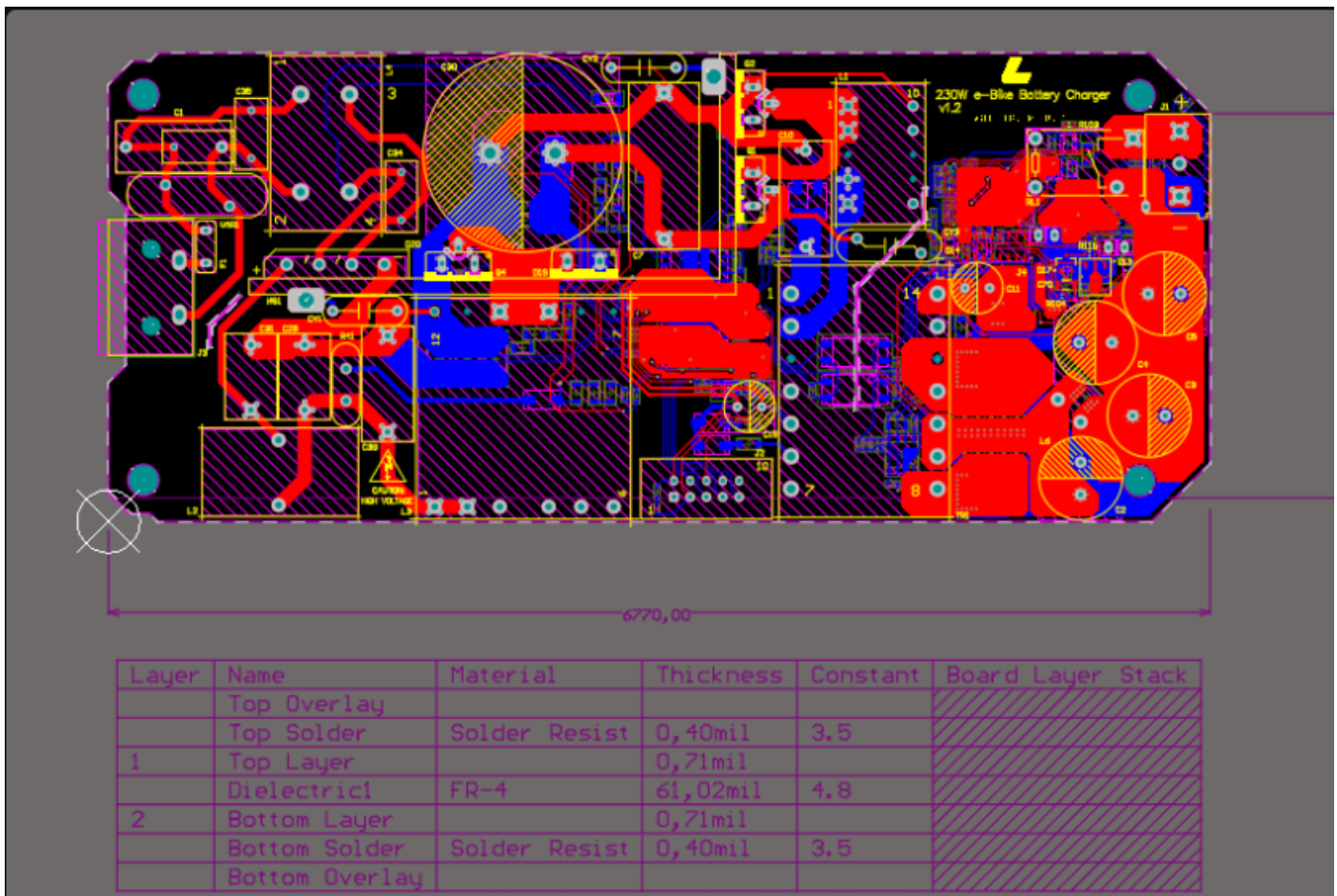


Figure 9: PCB design in two layers.

traces.

The output capacitors were placed directly after the SR stage to absorb current ripple and stabilize the battery output node. Wide copper pours were used on the secondary high-current path, and multiple vias were placed in parallel where needed to reduce current bottlenecks. The magnetic components were spaced so that airflow and thermal rise would not create hot spots in adjacent control circuitry.

7.0.5 Control and Low-Voltage Routing

The control board was isolated from the primary switching region and placed in a quieter area of the PCB. Analog sensing traces from the voltage dividers, current sensors, and temperature sensors were routed away from the half-bridge node and the PFC switching loop. ADC input traces were kept short and were filtered at the source using small RC networks where appropriate. Digital communication lines such as UART, I²C, and SWD were grouped and

routed with clear return paths.

The MCU decoupling capacitors were placed immediately next to the power pins, and the crystal or timing source was isolated from high-current traces to avoid jitter and noise injection. The low-voltage rails from the buck and LDO stages were distributed with local bypass capacitors at each IC. To improve robustness, test points were included on the main rails, gate-drive supplies, feedback nodes, and key enable signals.

7.0.6 Methods Used in Layout Verification

Several design-verification methods were used before releasing the layout:

- ERC and DRC checks were performed to identify schematic and layout violations.
- Net classes and clearance rules were used to enforce spacing by voltage domain.
- Polygon pour connectivity was verified to ensure low-



3D and Mechanical Design

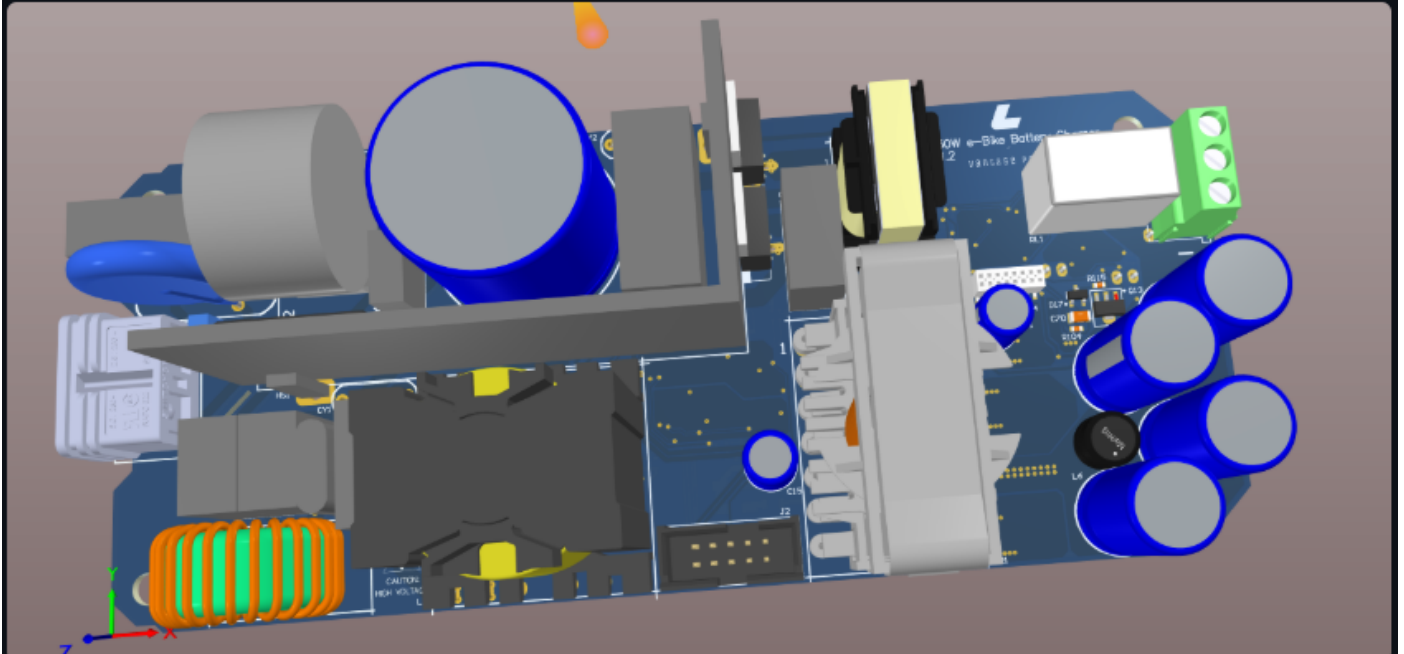


Figure 10: Three-dimensional view of the completed two-layer PCB layout.

impedance current return paths.

- 3D view inspection was used to confirm component height, spacing, and connector accessibility.
- High-current loop paths were reviewed to reduce EMI and optimize thermal spreading.
- Critical component placement was checked against the thermal model and expected airflow direction.

Figure 9 shows the final two-layer PCB layout, while Figure 10 provides a 3D view of the assembled board. The layout confirms that the power stage, control circuitry, and isolation domains were implemented within a compact footprint while preserving the spacing and routing discipline required for a high-voltage charger.

The schematic and PCB design process in Altium Designer followed a block-based and rule-driven methodology. The layout prioritized high-current loop minimization, thermal management, isolation spacing, and low-noise sensing, while the 3D review and design-rule checks provided an additional layer of verification before fabrication. This implementation strategy produced a manufacturable two-layer board suitable for prototype testing and system validation.

8 Embedded Firmware Development

The embedded firmware was developed in STM32CubeIDE for the STM32L0 microcontroller and organized into interrupt-driven control tasks, ADC acquisition routines, communication handlers, fault management, and PWM generation for the power stages. The pin assignment was defined early in the development process to ensure that the ADC channels, timer outputs, serial interfaces, and fault inputs were mapped consistently with the schematic and PCB layout.

8.0.1 Microcontroller Pin Assignment

The microcontroller pin map was configured to support analog sensing, frequency control of the LLC stage, synchronous rectifier control, communication interfaces, and protection inputs. Care was taken to keep the most noise-sensitive analog inputs away from switching outputs and to assign timer outputs with the required complementary and dead-time features.

8.0.2 Firmware Architecture

The firmware is structured as a real-time state machine with separate modules for startup, precharge, soft-start,



Table 1: STM32 microcontroller pin assignment used in STM32CubeIDE.

Signal	STM32 Pin	Peripheral Function	Description
Input voltage sense	PA0	ADC IN0	Scaled AC input voltage feedback
Input current sense	PA1	ADC IN1	Input current measurement for PFC control
PFC output voltage sense	PA2	ADC IN2	High-voltage DC bus feedback
Output current sense	PA3	ADC IN3	Battery charge current feedback
Temperature sense	PA4	ADC IN4	NTC or thermal monitoring input
LLC PWM high-side	PA8	TIM1_CH1	Complementary half-bridge drive output
LLC PWM low-side	PB13	TIM1_CH1N	Complementary half-bridge drive output
SR enable / control	PB0	GPIO Output	Synchronous rectifier control signal
Relay control	PB1	GPIO Output	Battery relay and precharge control
UART TX	PA9	USART1_TX	Debug and telemetry output
UART RX	PA10	USART1_RX	Debug and telemetry input
I ² C SCL	PB6	I2C1_SCL	Sensor or peripheral communication
I ² C SDA	PB7	I2C1_SDA	Sensor or peripheral communication
SWDIO	PA13	SWDIO	Programming and debugging interface
SWCLK	PA14	SWCLK	Programming and debugging interface
Fault input	PC13	GPIO Input	Hardware protection or external fault signal

constant-current charging, constant-voltage charging, fault handling, and shutdown. ADC sampling is performed either by timer-triggered conversions or periodic interrupts so that voltage and current measurements remain synchronized with the switching and control intervals. The LLC stage is regulated by frequency modulation rather than duty-cycle variation, while complementary PWM outputs are used to drive the primary half-bridge with fixed dead time.

8.0.3 Control Loop Operation

The firmware uses the measured input and output signals to compute control errors and update the operating point of the converter. Input voltage and current measurements are used for front-end supervision and PFC-related monitoring, while output voltage and current measurements determine the charging state. The LLC control loop adjusts switching frequency according to the measured error, so the duty cycle on the half-bridge remains fixed while the period is varied to regulate power transfer.

The general control relationship is:

$$e(t) = x_{\text{ref}} - x_{\text{meas}}$$

$$f_{\text{sw}} = f_{\text{nom}} + K_{\text{p}}e(t) + K_{\text{i}} \int e(t) dt$$

where f_{sw} is the commanded switching frequency and the PI controller determines the correction required to maintain the desired output voltage or current.

8.0.4 Programming and Debugging

The STM32CubeIDE environment was used for pin configuration, peripheral initialization, firmware generation, and debug access. The SWD interface was reserved for flashing and runtime debugging, while UART was used for diagnostics and parameter monitoring during bring-up. This approach simplifies validation of the sensing paths, PWM outputs, relay logic, and protection events before the full charger is energized. The embedded firmware layer provides the digital control infrastructure for the charger, with the STM32 microcontroller acting as the central control device. The pin assignment table defines the interface between the firmware and the hardware, ensuring that sensing, PWM generation, communication, and protection functions are mapped in a way that supports stable and deterministic operation.



9 PCB Assembly and Manufacturing

After the schematic capture, firmware bring-up planning, and PCB layout were completed, the final stage was manufacturing and assembly of the printed circuit board. This stage translated the design from CAD into physical hardware while preserving the electrical constraints required for high-voltage power conversion, thermal management, and safe testability. The fabrication workflow was designed with manufacturability, assembly repeatability, and debugging access in mind, using design-for-test (DFT), design-for-manufacturing (DFM), and design-for-assembly (DFA) principles throughout the process.

9.0.1 Manufacturing Workflow

The PCB was manufactured through JLCPCB, which provided both board fabrication and component assembly services for the selected design. The manufacturing package was prepared by exporting the final Gerber files, drill files, pick-and-place data, BOM, and assembly drawings from Altium Designer. These files were reviewed before submission to ensure that pad geometry, component orientation, polarity markings, and reference designators were consistent across the schematic, layout, and assembly documentation.

The fabrication stack-up and copper thickness were selected to support the current levels of the charger, while maintaining adequate spacing between primary and secondary domains. Special attention was given to solder mask clearance, silkscreen legibility, fiducial placement, and component footprint consistency to reduce the risk of assembly errors. The board outline, mounting holes, and connector positions were also verified against the mechanical enclosure before release.

9.0.2 Design for Manufacturability

DFM was applied to ensure that the PCB could be fabricated and assembled with high yield and low risk of rework. This included the use of standard component footprints, sufficient pad-to-pad spacing, and conservative trace widths in high-voltage and high-current areas. The layout avoided unnecessarily dense routing in regions that would complicate soldering or inspection, especially around the power stage, transformer, and control circuitry.

High-voltage design rules were respected during fabrication planning, including creepage and clearance constraints between live conductors and isolated domains. Copper pours were shaped to avoid sharp corners and narrow neck-downs that could create current bottlenecks or reduce process robustness. The design also accounted for solder mask expansion and component courtyard dimensions so that pick-and-place assembly could proceed without interference.

9.0.3 Design for Assembly

DFA considerations were incorporated to make the board easier to populate, inspect, and rework. Components with polarity sensitivity, such as diodes, electrolytic capacitors, ICs, and connectors, were placed with clear orientation markers and consistent reference directions. Large power components were grouped logically to reduce placement complexity and to simplify thermal and electrical debugging after assembly.

The component library was standardized so that footprints, pad sizes, and recommended land patterns matched the intended package types. This reduced the likelihood of footprint mismatches or tombstoning during reflow. Where possible, parts were selected in packages that are compatible with automated assembly and rework, such as QFN, SOIC, SOT-23, and 2512/1206 resistor and capacitor packages. Larger through-hole or mechanically stressed components, such as relays, inductors, and connectors, were positioned to improve solder joint strength and mechanical reliability.

9.0.4 Design for Test

DFT was implemented through a modular board structure and the inclusion of test points on all critical nodes. The design was partitioned into functional blocks, allowing input front-end, control logic, auxiliary supplies, and power-stage interfaces to be validated independently during bring-up. Test points were added for the main power rails, feedback signals, PWM outputs, fault lines, enable pins, and communication interfaces so that voltage and timing can be verified without probing directly on fine-pitch device pins.

The test points were placed with sufficient spacing for oscilloscope probes and multimeter clips, and they were labeled clearly in the silkscreen and documentation. This improves debugging efficiency and reduces the chance of



accidental short circuits during energized testing. In addition, the modular structure allows individual subsystems to be isolated for troubleshooting, which is especially useful when validating the PFC stage, LLC converter, synchronous rectification, and control firmware separately.

9.0.5 Component Placement and Assembly Considerations

The assembly process was guided by the electrical and thermal function of each part. High-power components such as MOSFETs, rectifiers, inductors, transformers, and current-sense elements were grouped to minimize current loop area and to support heat spreading. Control ICs and sensing components were placed in quieter regions of the board to reduce interference from switching nodes and high di/dt paths.

Decoupling capacitors were placed as close as possible to the supply pins of each IC to minimize supply impedance and improve transient response. The placement of the microcontroller, gate drivers, and current-sense amplifiers was chosen to keep signal traces short and clean. Passive components were oriented consistently to ease inspection, and sufficient spacing was kept between hot components to avoid thermal coupling and to improve airflow.

9.0.6 Assembly Verification

After assembly, the board was inspected for solder quality, alignment, polarity correctness, and continuity before power-up. Initial verification was performed with a continuity test and low-voltage checks on the auxiliary rails. The input stage, low-voltage supplies, and control board were then tested independently before energizing the full power stage. This staged verification approach reduces the risk of catastrophic failure and allows faults to be identified early.

The test strategy also included checking the response of the relay control, precharge sequence, and fault-handling logic before applying full charging power. Once the low-voltage circuitry was confirmed, the converter could be energized gradually with monitoring of input current, bus voltage, switching waveforms, and output behavior. This method supports safe debugging and limits the possibility of damage during the first power-up.

The PCB manufacturing and assembly stage was executed with DFM, DFA, and DFT principles integrated into the design process. JLCPCB was used for board fabrica-

tion and component placement, while modular layout and test-point access ensured that the charger could be assembled and validated in a structured manner. This approach improved manufacturability, reduced assembly risk, and supported efficient prototype bring-up and troubleshooting.

10 Prototype Testing and Validation

Prototype testing was performed after PCB assembly to verify correct operation of each subsystem and to confirm that the charger met the intended electrical, thermal, and functional requirements. The validation process was intentionally staged so that the design could be brought up safely, starting with low-voltage checks and progressing gradually to full-power operation. This method reduced the risk of hardware damage and made it possible to isolate faults quickly if any subsystem behaved unexpectedly.

10.0.1 Testing Sequence

The testing sequence began with continuity inspection and visual verification of the assembled PCB. After confirming solder quality, component orientation, and the absence of shorts on the low-voltage rails, the auxiliary supply was powered independently and checked for correct output voltages. The control board was then powered and the firmware was loaded to verify MCU startup, timer operation, ADC readings, relay control, and communication interfaces.

Once the low-voltage circuitry had been validated, the input power stage was energized with reduced power and monitored carefully. Measurements were taken at multiple locations across the converter, including the AC input, rectified DC bus, primary switching stage, secondary output, and battery connection point. This allowed the design to be verified stage by stage, from input power quality and bus stability to transformer transfer behavior, rectifier performance, output regulation, and battery charging response.

During constant-current charging, the charger maintains a near-constant output current while the battery voltage rises gradually. As the battery approaches its target voltage, the system transitions into constant-voltage mode, where the voltage is held steady and the current naturally tapers down until the battery is nearly full. This behavior confirms that the firmware control loop, sensing chain, and



power stage are working together as intended.



Figure 11: Laboratory bench setup used for prototype testing.

10.0.2 Electrical Validation

Electrical validation focused on the voltage and current waveforms at each important node in the power chain. The AC input was checked to confirm correct rectification and PFC bus formation. The DC bus was then monitored for ripple, regulation, and stability under load. The LLC primary switching node was inspected to confirm resonant operation and to verify that the switching transitions remained consistent with the simulation results.

On the secondary side, output voltage and current were recorded during both constant-current and constant-voltage charging. The transition between charging modes was used as a functional indicator of correct closed-loop control. The battery connection point was also checked to ensure that relay operation, precharge timing, and charger enable logic were functioning correctly during startup and shutdown.

10.0.3 Thermal Validation

Thermal measurements were taken to identify hot spots across the primary switches, transformer, synchronous rectifier, and auxiliary power components during load testing. This information was used to evaluate loss distribution and confirm that the layout and cooling strategy were suitable for continuous operation. The thermal survey also highlighted components that may require revised copper area, airflow management, or package selection in the next hardware revision.

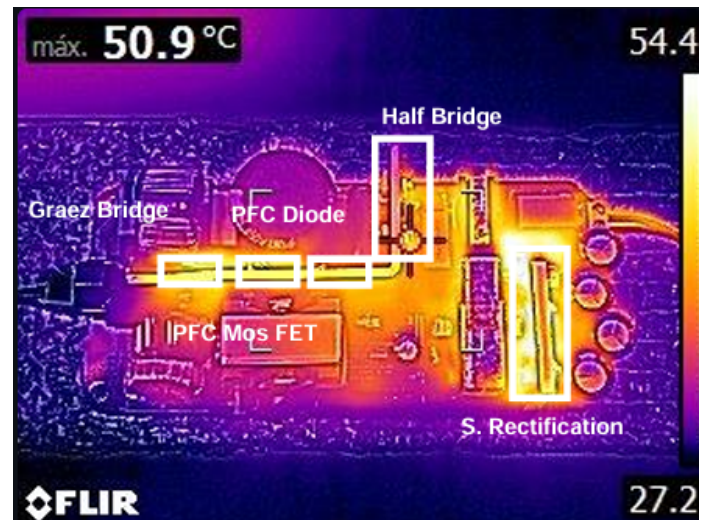


Figure 12: Thermal measurements at 24 V output and 5 A load, with 25°C ambient temperature.

10.0.4 Functional Results

The prototype results confirmed the overall operating principle of the charger. The constant-current phase provided stable charging current as the battery voltage increased, and the constant-voltage phase reduced the current naturally near the top of charge. This validates both the sensing architecture and the firmware logic used to transition between charging states.

In addition, the thermal image indicates the distribution of heat among the major power components, which is important for assessing long-term reliability. Areas with elevated temperature can be linked to conduction loss, switching loss, or insufficient copper spreading, while cooler regions indicate effective heat distribution and acceptable device loading. These results provide direct feedback for possible optimization in a future PCB revision.

Prototype testing and validation demonstrated that the charger could be brought up safely in stages and that the major functional blocks operated correctly under laboratory conditions. Electrical measurements confirmed stable converter behavior across the input, DC bus, primary stage, and output stage, while thermal testing provided insight into loss distribution and cooling performance. Together, these results show that the design is functionally sound and ready for further refinement or extended endurance testing.

Figure 13 shows the charger installed in the final e-bike prototype, where the PCB modules are clearly visible during validation testing. This setup demonstrates the mechanical integration of the charger and confirms that the

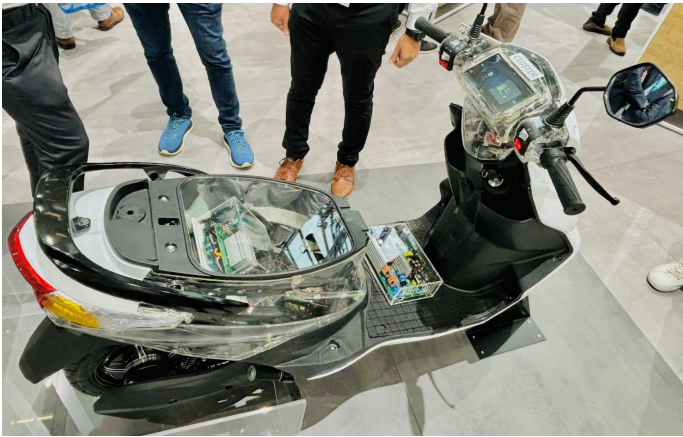


Figure 13: The charger integrated into the final e-bike prototype, with the charger PCBs clearly visible during validation testing.

design is suitable for real-world use in the full system.

11 Data Analysis and Test Automation

The final stage of the design process focused on data analysis and test automation, which were used to support prototype validation, improve measurement repeatability, and accelerate debugging during bring-up. As the hardware was tested across multiple operating points, manual logging became increasingly impractical, especially when comparing charging behavior, thermal performance, and converter waveforms over repeated experiments. To address this, Python was used as a lightweight automation layer between the laboratory instrumentation and the analysis workflow.

11.0.1 Test Automation Framework

Python scripts were developed to automate test capture, organize measurement data, and standardize result processing. The scripts were used alongside bench instruments such as oscilloscopes, multimeters, temperature probes, and power supplies to record electrical and thermal measurements in a structured format. This reduced operator workload during testing and improved consistency between measurement runs, particularly when the same test sequence needed to be repeated at different load conditions, input voltages, or battery states.

The automation workflow also made it easier to synchronize measurements from multiple sources. For example, voltage, current, and temperature data could be captured during the same test interval and then associated with a

specific operating point. This was especially useful during converter bring-up, where small changes in duty cycle, load level, or charging mode could significantly affect performance. By organizing the measurements in a repeatable format, the test process became more efficient and easier to review.

11.0.2 Measurement Processing

The collected data was then processed to extract trends and evaluate the behavior of the charger under different conditions. Python was used to clean raw measurement files, align datasets from different instruments, and generate plots for design review. This allowed the measured results to be compared directly against simulation outputs and expected design targets.

Typical analysis tasks included waveform inspection, average and ripple calculations, thermal trend analysis, and charging profile evaluation. Voltage and current waveforms were used to confirm correct converter operation and to identify transient effects during mode transitions. Thermal data was analyzed to locate hot spots and assess whether the layout and cooling approach were adequate for continuous operation. In addition, charging curves were reviewed to verify the transition between constant-current and constant-voltage modes.

11.0.3 Example Test Activities

The automation scripts supported several recurring test activities throughout development:

- Capture voltage and current waveforms at the AC input, DC bus, switching stage, output stage, and battery connection point.
- Log thermal data from the primary power devices, transformer, rectifier stage, and auxiliary supply components.
- Compare simulated and measured converter behavior across load conditions and charging states.
- Generate plots and summary figures for lab documentation, report preparation, and design review.
- Track repeated test runs to identify variation between measurements and highlight unstable operating conditions.



11.0.4 Benefits to Validation

Using Python for test automation improved the overall validation workflow in several ways. First, it reduced the time required to collect and organize data during iterative testing. Second, it improved repeatability by ensuring that test conditions, file naming, and analysis steps were applied consistently. Third, it made debugging faster by allowing the measured behavior of individual subsystems to be reviewed immediately after each test run.

This approach was particularly valuable for a mixed-signal power electronics design, where electrical performance, thermal behavior, and firmware response must all be evaluated together. Automated analysis made it possible to correlate these domains and identify design limitations more efficiently. As a result, the validation process not only confirmed correct system operation but also provided actionable feedback for future hardware revisions and firmware improvements.

Data analysis and test automation formed the final layer of the development workflow. Together, they transformed raw lab measurements into structured evidence of performance, supported comparison between simulation and hardware results, and improved the efficiency of prototype validation. In combination with the staged testing strategy used earlier, this automation framework helped ensure that the charger was evaluated systematically and documented thoroughly.

12 Future Improvements

Although the current prototype successfully demonstrates the main charging functionality, several areas have been identified for improvement in the next hardware revision. These enhancements focus on improving efficiency, thermal performance, electromagnetic compatibility, system observability, and overall robustness of the design.

- Replace the primary MOSFETs with GaN devices to reduce switching losses, improve high-frequency performance, and increase overall conversion efficiency.
- Further optimize light-load efficiency by refining control strategies, reducing standby losses, and improving the behavior of the power stage at lower output power levels.

- Improve thermal spreading by revising the copper layout, enlarging heat dissipation areas, and optimizing airflow paths around the hottest components.
- Expand BMS communication features to support richer battery-state information, improved fault reporting, and more intelligent coordination with the charger.
- Refine EMI performance through layout optimization, better return-path control, and additional filtering at the input and switching stages.
- Add more diagnostic and telemetry functions in firmware, including additional fault logging, operating-state reporting, and live monitoring of key electrical variables.

These improvements would make the next revision more efficient, easier to debug, and better suited for long-term operation in a practical e-bike charging application.

13 Software Tools

Several software tools were used throughout the design process to support circuit simulation, PCB development, firmware implementation, and experimental validation. Each tool served a specific role in the workflow and contributed to a structured transition from concept to working prototype.

- LTspice was used for circuit simulation and preliminary verification of key analog and power-stage behavior.
- Altium Designer was used for schematic capture, PCB layout, footprint management, and generation of manufacturing files.
- STM32CubeIDE was used for embedded firmware development, debugging, and microcontroller configuration.
- Python was used for test automation, data logging, and post-processing of measurement results.

Together, these tools supported the full development cycle, from early-stage simulation and hardware design to prototype bring-up and validation.

14 Project Summary

This project developed and validated a high-performance e-bike charger from concept to prototype. The work cov-



ered system definition, circuit simulation, firmware development, PCB design, manufacturing, assembly, laboratory testing, and post-processing of results, resulting in a complete hardware platform that demonstrates the intended charging functionality.

The design process began with the definition of electrical and functional requirements, including charging control, power conversion, measurement, protection, and communication. Based on these requirements, the charger architecture was divided into major functional blocks and evaluated through simulation before hardware implementation. This helped identify key design constraints early and guided component selection, control strategy development, and topology refinement.

Firmware development formed an essential part of the project, since the charger relies on coordinated control of sensing, switching, protection, and charging state management. The embedded software was implemented to support startup sequencing, operating-mode transitions, and monitoring of electrical variables. This ensured that the hardware could be exercised safely during bring-up and that the charging process behaved as intended across different operating conditions.

The PCB was designed with manufacturability, assembly, and testability in mind. DFM and DFA principles were applied to improve fabrication yield and simplify assembly, while DFT features such as modular structure and test points supported debugging and verification. The board was then fabricated and assembled through JLCPCB, enabling a rapid transition from digital design to physical prototype.

Prototype testing and validation confirmed correct operation of the major subsystems. Measurements were taken across the input stage, DC bus, switching stage, output stage, and battery interface to verify converter behavior step by step. Thermal testing identified the most critical heat-generating components and provided feedback for future optimization. In parallel, automated data logging and analysis in Python improved repeatability and made it easier to compare simulation results with measured performance.

Overall, the project demonstrates a complete engineering workflow for developing an advanced e-bike charging system. It shows how simulation, embedded control, PCB

design, prototype validation, and data-driven analysis can be combined into a structured development process. The result is a functional prototype and a clear path toward future improvements in efficiency, thermal performance, EMI behavior, and diagnostic capability.

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A Appendix

A.1 Embedded Firmware

Listing 1: STM32 HAL-based firmware

```
/* USER CODE BEGIN Header */
/*
 * Project: 360W LLC Battery Charger
 * Target: STM32L0 series
 * Toolchain: STM32CubeIDE / HAL
 * Description :
 *   - ADC sensing for input/output voltage and current
 *   - Frequency-controlled LLC primary stage
 *   - Synchronous rectifier coordination
 *   - Relay/precharge control
 *   - UART diagnostics
 *   - I2C peripheral support
 */
/* USER CODE END Header */

#include "main.h"
#include <cmath>
#include <cstring>

/* External handles generated by CubeMX */
ADC_HandleTypeDef hadc;
TIM_HandleTypeDef htim1;
UART_HandleTypeDef huart1;
I2C_HandleTypeDef hi2c1;

/* Application constants */
static constexpr float VREF = 3.3f;
static constexpr uint16_t ADC_MAX = 4095;
static constexpr float VIN_DIV_RATIO = 100.0f;
static constexpr float VBUS_DIV_RATIO = 120.0f;
static constexpr float IOUT_GAIN = 100.0f;
static constexpr float ADC_DT = 0.001f;
static constexpr float FREQ_MIN_HZ = 60000.0f;
static constexpr float FREQ_MAX_HZ = 180000.0f;
static constexpr float FREQ_NOM_HZ = 100000.0f;

/* Setpoints */
static constexpr float VOUT_REF = 24.0f;
static constexpr float IOUT_REF_CC = 15.0f;
```



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```
/* PI gains for frequency control */
static constexpr float KP_V          = 2500.0f;
static constexpr float KI_V          = 180.0f;
static constexpr float KP_I          = 1800.0f;
static constexpr float KI_I          = 120.0f;

/* ADC channels / measured values */
volatile uint16_t adc_raw_vin  = 0;
volatile uint16_t adc_raw_iin  = 0;
volatile uint16_t adc_raw_vbus = 0;
volatile uint16_t adc_raw_iout = 0;
volatile uint16_t adc_raw_ntc  = 0;

volatile float vin_meas  = 0.0f;
volatile float iin_meas  = 0.0f;
volatile float vbus_meas = 0.0f;
volatile float iout_meas = 0.0f;
volatile float temp_meas = 0.0f;

/* Control state */
enum class ChargerState
{
    IDLE = 0,
    PRECHARGE,
    SOFT_START,
    CONSTANT_CURRENT,
    CONSTANT_VOLTAGE,
    FAULT
};

volatile ChargerState state = ChargerState::IDLE;
volatile bool fault_active = false;
volatile bool bms_permit = false;

/* PI controller */
struct PIController
{
    float kp;
    float ki;
    float integrator;
    float out_min;
    float out_max;
};
```



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```
PIController pi_v { KP_V, KI_V, 0.0f, FREQ_MIN_HZ, FREQ_MAX_HZ };
PIController pi_i { KP_I, KI_I, 0.0f, FREQ_MIN_HZ, FREQ_MAX_HZ };

/* Function prototypes */
void SystemClock_Config(void);
static void MX_GPIO_Init(void);
static void MX_ADC_Init(void);
static void MX_TIM1_Init(void);
static void MX_USART1_UART_Init(void);
static void MX_I2C1_Init(void);

static void Error_Handler(void);
static void Log(const char* msg);
static float ADC_ToVoltage(uint16_t raw, float ratio);
static float ADC_ToCurrent(uint16_t raw, float gain);
static float ADC_ToTemperature(uint16_t raw);

static float PI_Update(PIController& pi, float ref, float meas, float dt);
static void UpdateLLCFrequency(float freq_hz);
static void StartPrecharge(void);
static void CloseMainRelay(void);
static void OpenMainRelay(void);
static void SetSR(bool enable);
static void EnterFault(const char* reason);
static void ControlTask(void);
static void ReadADCValues(void);
static void CheckProtections(void);

int main(void)
{
    HAL_Init();
    SystemClock_Config();

    MX_GPIO_Init();
    MX_ADC_Init();
    MX_TIM1_Init();
    MX_USART1_UART_Init();
    MX_I2C1_Init();

    HAL_TIM_PWM_Start(&htim1, TIM_CHANNEL_1);
    HAL_TIM_PWM_Start(&htim1, TIM_CHANNEL_2);

    SetSR(false);
    OpenMainRelay();
}
```



```
UpdateLLCFrequency(FREQ_NOM_HZ);

Log("System boot\r\n");

while (1)
{
    ReadADCValues();
    CheckProtections();
    ControlTask();
    HAL_Delay(1);
}

static void ReadADCValues(void)
{
    HAL_ADC_Start(&hadc);

    HAL_ADC_PollForConversion(&hadc, 10);
    adc_raw_vin = HAL_ADC_GetValue(&hadc);

    HAL_ADC_PollForConversion(&hadc, 10);
    adc_raw_iin = HAL_ADC_GetValue(&hadc);

    HAL_ADC_PollForConversion(&hadc, 10);
    adc_raw_vbus = HAL_ADC_GetValue(&hadc);

    HAL_ADC_PollForConversion(&hadc, 10);
    adc_raw_iout = HAL_ADC_GetValue(&hadc);

    HAL_ADC_PollForConversion(&hadc, 10);
    adc_raw_ntc = HAL_ADC_GetValue(&hadc);

    HAL_ADC_Stop(&hadc);

    vin_meas = ADC_ToVoltage(adc_raw_vin, VIN_DIV_RATIO);
    vbus_meas = ADC_ToVoltage(adc_raw_vbus, VBUS_DIV_RATIO);
    iin_meas = ADC_ToCurrent(adc_raw_iin, 0.25f);
    iout_meas = ADC_ToCurrent(adc_raw_iout, 0.02f);
    temp_meas = ADC_ToTemperature(adc_raw_ntc);
}

static void ControlTask(void)
{
    if (fault_active)
```



```
{
    state = ChargerState::FAULT;
}

switch (state)
{
    case ChargerState::IDLE:
        OpenMainRelay();
        SetSR(false);
        UpdateLLCFrequency(FREQ_NOM_HZ);
        if (bms_permit && vbus_meas > 20.0f)
        {
            state = ChargerState::PRECHARGE;
            Log("State: PRECHARGE\r\n");
        }
        break;

    case ChargerState::PRECHARGE:
        StartPrecharge();
        if (std::fabs(vbus_meas - VOUT_REF) < 2.0f)
        {
            CloseMainRelay();
            SetSR(true);
            state = ChargerState::SOFT_START;
            Log("State: SOFT_START\r\n");
        }
        break;

    case ChargerState::SOFT_START:
    {
        float f = PI_Update(pi_v, VOUT_REF, vbus_meas, ADC_DT);
        UpdateLLCFrequency(f);
        if (std::fabs(vbus_meas - VOUT_REF) < 1.0f)
        {
            state = ChargerState::CONSTANT_CURRENT;
            Log("State: CONSTANT_CURRENT\r\n");
        }
        break;
    }

    case ChargerState::CONSTANT_CURRENT:
    {
        float f = PI_Update(pi_i, IOUT_REF_CC, iout_meas, ADC_DT);
        UpdateLLCFrequency(f);
    }
}
```




```
        if (iout_meas < IOUT_REF_CC && vbus_meas > VOUT_REF - 0.5f)
        {
            state = ChargerState::CONSTANT_VOLTAGE;
            Log("State: CONSTANT_VOLTAGE\r\n");
        }
        break;
    }

    case ChargerState::CONSTANT_VOLTAGE:
    {
        float f = PI_Update(pi_v, VOUT_REF, vbus_meas, ADC_DT);
        UpdateLLCFrequency(f);
        break;
    }

    case ChargerState::FAULT:
    {
        OpenMainRelay();
        SetSR(false);
        UpdateLLCFrequency(FREQ_MAX_HZ);
        break;
    }

    default:
    {
        state = ChargerState::FAULT;
        break;
    }
}

static float PI_Update(PIController& pi, float ref, float meas, float dt)
{
    float err = ref - meas;
    pi.integrator += err * pi.ki * dt;
    if (pi.integrator > pi.out_max) pi.integrator = pi.out_max;
    if (pi.integrator < pi.out_min) pi.integrator = pi.out_min;

    float out = pi.kp * err + pi.integrator;
    if (out > pi.out_max) out = pi.out_max;
    if (out < pi.out_min) out = pi.out_min;
    return out;
}

static void UpdateLLCFrequency(float freq_hz)
{
    if (freq_hz < FREQ_MIN_HZ) freq_hz = FREQ_MIN_HZ;
    if (freq_hz > FREQ_MAX_HZ) freq_hz = FREQ_MAX_HZ;
}
```



```
uint32_t timer_clk = HAL_RCC_GetPCLK2Freq();
uint32_t arr = static_cast<uint32_t>((timer_clk / freq_hz) - 1.0f);

__HAL_TIM_SET_AUTORELOAD(&htim1, arr);
__HAL_TIM_SET_COMPARE(&htim1, TIM_CHANNEL_1, arr / 2);
__HAL_TIM_SET_COMPARE(&htim1, TIM_CHANNEL_2, arr / 2);
}

static void StartPrecharge(void)
{
    HAL_GPIO_WritePin(PRECHARGE_GPIO_Port, PRECHARGE_Pin, GPIO_PIN_SET);
    HAL_GPIO_WritePin(MAIN_RELAY_GPIO_Port, MAIN_RELAY_Pin, GPIO_PIN_RESET);
}

static void CloseMainRelay(void)
{
    HAL_GPIO_WritePin(MAIN_RELAY_GPIO_Port, MAIN_RELAY_Pin, GPIO_PIN_SET);
    HAL_Delay(50);
    HAL_GPIO_WritePin(PRECHARGE_GPIO_Port, PRECHARGE_Pin, GPIO_PIN_RESET);
}

static void OpenMainRelay(void)
{
    HAL_GPIO_WritePin(MAIN_RELAY_GPIO_Port, MAIN_RELAY_Pin, GPIO_PIN_RESET);
    HAL_GPIO_WritePin(PRECHARGE_GPIO_Port, PRECHARGE_Pin, GPIO_PIN_RESET);
}

static void SetSR(bool enable)
{
    HAL_GPIO_WritePin(SR_EN_GPIO_Port, SR_EN_Pin, enable ? GPIO_PIN_SET : GPIO_PIN_RESET);
}

static void CheckProtections(void)
{
    if (vbus_meas > 30.0f || iout_meas > 16.0f || temp_meas > 80.0f)
    {
        EnterFault("Protection trip");
    }

    if (HAL_GPIO_ReadPin(BMS_OK_GPIO_Port, BMS_OK_Pin) == GPIO_PIN_RESET)
    {
        EnterFault("BMS not permitted");
    }
}
```



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```
}

static void EnterFault(const char* reason)
{
    fault_active = true;
    state = ChargerState::FAULT;
    OpenMainRelay();
    SetSR(false);
    Log(reason);
    Log("\r\n");
}

static float ADC_ToVoltage(uint16_t raw, float ratio)
{
    float v = (static_cast<float>(raw) * VREF) / ADC_MAX;
    return v * ratio;
}

static float ADC_ToCurrent(uint16_t raw, float gain)
{
    float v = (static_cast<float>(raw) * VREF) / ADC_MAX;
    return v * gain;
}

static float ADC_ToTemperature(uint16_t raw)
{
    float v = (static_cast<float>(raw) * VREF) / ADC_MAX;
    return v * 25.0f;
}

static void Log(const char* msg)
{
    HAL_UART_Transmit(&huart1, reinterpret_cast<uint8_t*>(const_cast<char*>(msg)), std::strlen(msg), 1000);
}

void HAL_GPIO_EXTI_Callback(uint16_t GPIO_Pin)
{
    if (GPIO_Pin == FAULT_IN_Pin)
    {
        EnterFault("External fault interrupt");
    }
}

void HAL_TIM_PeriodElapsedCallback(TIM_HandleTypeDef* htim)
```



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```
{
    if (htim->Instance == TIM1)
    {
        ControlTask();
    }
}

static void Error_Handler(void)
{
    __disable_irq();
    while (1)
    {
        HAL_GPIO_TogglePin(LED_GPIO_Port, LED_Pin);
        HAL_Delay(100);
    }
}

/* Peripheral init functions are generated by STM32CubeIDE and omitted here. */
/* SystemClock_Config(), MX_GPIO_Init(), MX_ADC_Init(), MX_TIM1_Init(),
   MX_USART1_UART_Init(), MX_I2C1_Init() */
```



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A.2 Bill of Materials (BOM)



Table 2: Summary of selected vital components for the charger design.

Component	Specification	Recommended Part
Input bridge rectifier	600–1000 V, 8–10 A, low forward loss, high surge current capability	KBPC1010 or equivalent
Boost PFC MOSFET	650 V, low $R_{DS(on)}$, fast switching, TO-247 package	Infineon / onsemi 650 V super-junction MOSFET
PFC diode	650 V ultrafast or SiC diode, low reverse recovery	650 V SiC diode class part
PFC controller	Universal input boost PFC controller with current-mode control	TI PFC controller class part
Bulk capacitor	450 V electrolytic, low ESR, high ripple current, $\geq 360 \mu\text{F}$ total	$2 \times 220 \mu\text{F}$, 450 V electrolytic
Primary LLC MOSFETs	650 V, low Q_g , low C_{oss} , TO-247 package	650 V super-junction MOSFETs
LLC resonant capacitor	Polypropylene film capacitor, high RMS current, low ESR	100 nF film capacitor class
Transformer	Ferrite core, reinforced insulation, approx. 8:1 turns ratio	Custom-wound ferrite transformer
Secondary SR MOSFETs	60 V, very low $R_{DS(on)}$, high current handling	60 V LFPK56 MOSFET
SR controller	Adaptive or drain-sense synchronous rectifier controller	TI UCC24624
Microcontroller	Low-power MCU with ADCs, timers, UART, I ² C, SWD	STM32L053C8T6
Gate driver	Half-bridge driver with UVLO, high dv/dt immunity	TI half-bridge gate-driver class part
Buck regulator	High-efficiency synchronous buck for control rails	TI TPSM63603 or TPS54202 class
LDO regulator	Low-noise regulator for MCU and analog rails	TI TPS7A44 class
Battery relay / contactor	DC-rated relay or contactor, low contact resistance, 30 A class or higher	DC power relay / contactor
Precharge resistor	High-pulse wirewound resistor, suitable for in-rush limiting	56 Ω to 68 Ω , 25 W pulse class
Current sense resistor	2 m Ω , 5 W precision shunt, low inductance	Vishay WHPT10 or equivalent
Voltage sense resistors	1% high-voltage resistor string, 200 V per resistor minimum	99 k Ω /1 k Ω and 119 k Ω /1 k Ω dividers
NTC sensor	Temperature monitoring on heatsink or transformer	10 k Ω NTC thermistor